

The Supra-Hodge Laplacian: A Unified Spectral Operator for Multi-View Higher-Order Relational Systems

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Abstract

We introduce the *Supra-Hodge Laplacian*, a new spectral operator that unifies two previously orthogonal generalizations of the classical graph Laplacian: the supra-Laplacian, which encodes cross-layer coupling in multilayer graph systems, and the combinatorial Hodge Laplacian, which encodes higher-order topological structure within a single simplicial complex. Fixing a family of complexes on a shared vertex set (one view per layer), an aligned reference set of p -simplices, and nonnegative scalar couplings between layers, the order- p Supra-Hodge Laplacian L_p is the block operator on stacked ambient p -chains that augments embedded layerwise Hodge Laplacians $H_p^{(i)}$ with a Kronecker-type coupling $L_{\text{inter}} \otimes I_{N_p}$ from the inter-layer graph. For a stacked signal, its quadratic energy splits into within-layer topological smoothness and cross-layer disagreement. The central formal result is the *Consensus-Harmonic Nullspace Theorem*: $\ker(L_p)$ is exactly the set of stacked signals that are simultaneously harmonic in every layer and mutually consensual across every connected component of the inter-layer coupling graph.

The construction is provably consistent: in the single-layer, order-0, no-higher-order limit it recovers the standard graph Laplacian; at $p = 0$ it reduces to the classical supra-Laplacian; and at $k = 1$ it reduces to the single-layer Hodge Laplacian. Spectral analysis yields a generalized Fiedler theory for coupled higher-order systems, with low-energy eigenmodes identifying coherent multi-view harmonic structure and conflict-frontier modes identifying topological inconsistency across views. We develop a low-energy projector $P_{p,r}$ that achieves principled spectral compression of multi-view chain signals. An application to multi-view large-language-model reasoning formalizes the operator as a diagnostic and compressive device over structured reasoning traces. Proof-of-concept experiments on synthetic relational benchmarks support the theoretical predictions and show quantitative advantages over pairwise and single-view spectral baselines.

Keywords: Hodge Laplacian, supra-Laplacian, simplicial complexes, multilayer graphs, spectral methods, higher-order topology, multi-view learning, LLM reasoning, spectral compression.

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Notation Summary

Symbol	Meaning
$K^{(i)}$	Simplicial complex for layer i
$H_p^{(i)}$	Layerwise Hodge Laplacian on the aligned ambient space $\tilde{C}_p$
N_p	Ambient p -chain dimension, $N_p = \Sigma_p $
$\tilde{C}_p$	Common aligned ambient p -chain space, $\tilde{C}_p \cong \mathbb{R}^{N_p}$
L_{inter}	Inter-layer graph Laplacian
L_p	Supra-Hodge Laplacian
$v^{(i)}$	p -chain signal in layer i
Σ_p	Global simplex index set

1 Introduction

1.1 Motivation: Multi-View and Higher-Order Relational Systems

Modern relational systems are neither pairwise nor single-view. A protein-interaction network simultaneously encodes physical binding, regulatory co-expression, and genetic interaction, each of which can form higher-order complexes involving three or more proteins jointly. A multi-hop reasoning agent maintains semantic, evidential, causal, and temporal layers over the same set of working entities, and within each layer certain groups of three entities form genuine triplets that cannot be decomposed into their pairwise parts. A knowledge graph for scientific planning contains both citation layers and experimental-dependency layers, each populated with cliques and triangles that carry structurally distinct information.

Two powerful spectral frameworks have been developed to address these complexities individually, but neither addresses both at once. The *supra-Laplacian* framework [2, 3] couples multiple graph layers over a shared vertex set via a block operator, enabling multi-view spectral analysis of pairwise relations. The *combinatorial Hodge Laplacian* [4–6] generalizes the graph Laplacian to simplicial complexes, enabling spectral analysis of higher-order interactions within a single relational view. Neither framework, in its usual form, is aimed at the full combination: multiple views, each possessing higher-order structure, coupled across views.

We construct the *Supra-Hodge Laplacian*, a block operator that is simultaneously a supra-Laplacian across views and a Hodge Laplacian within each view. The primary contribution of this paper is mathematical; the reasoning application in Section 13 serves to illustrate the operator’s relevance to structured multi-view inference systems.

1.2 Why Pairwise Graph Methods Are Insufficient

Consider three agents forming a stable epistemic coalition whose stability is a genuinely three-body phenomenon: remove any single edge and the coalition dissolves, yet the pairwise adjacency between any pair is nonzero. A graph Laplacian encodes only the pairwise structure; it cannot represent the qualitatively different status of the filled triangle versus the hollow cycle of three edges. This is not a quirk: it is the defining limitation of all pairwise methods.

Within a single Hodge framework, this limitation is overcome for one relational view. But when k views exist—each with its own triangles, its own harmonic cycles, its own curl-free and gradient-free signals—a single complex conflates their topologies. Analyzing each view in isolation misses cross-view interactions; analyzing all views in a single complex erases the boundaries between views.

The supra-Laplacian handles cross-view interactions, but only at the level of vertices and pairwise edges. It cannot measure higher-order inconsistency within a view, nor can it detect the collapse of a harmonic cycle in one view due to pressure from a topologically simpler view coupled to it.

1.3 What This Paper Contributes

This paper makes the following contributions.

1. **Operator construction.** We define the order- p Supra-Hodge Laplacian L_p in Section 9 as a block operator on a common aligned p -chain space, combining embedded layerwise Hodge Laplacians $H_p^{(i)}$ with an inter-layer coupling term induced by the layer graph Laplacian L_{inter} .
2. **Main theorem.** We prove the Consensus-Harmonic Nullspace Theorem (Theorem 10.7), giving a complete algebraic characterization of $\ker(L_p)$.

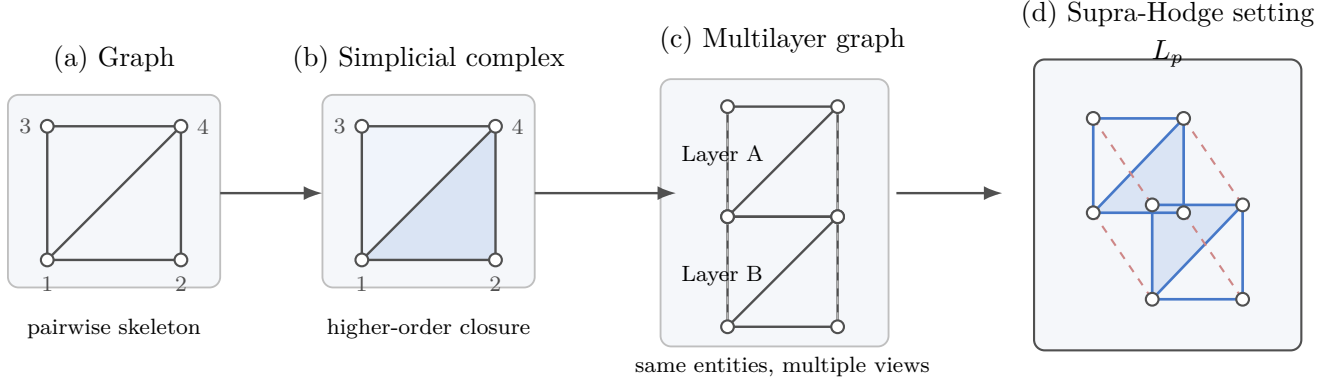


Figure 1: Progressive emergence of the Supra-Hodge setting: a pairwise graph becomes a simplicial complex by adding higher-order faces, then a multilayer graph by duplicating the entity set across views, and finally a coupled higher-order system in which intra-layer topology and inter-layer coupling coexist inside a single operator model.

3. **Reduction propositions.** We prove that L_p reduces to the classical supra-Laplacian at $p = 0$ and to the single-layer Hodge Laplacian at $k = 1$ (Propositions 10.4 and 10.5).
4. **Spectral theory.** We develop a generalized Fiedler theory and formalize the low-energy projector $P_{p,r}$ for spectral compression (Section 11).
5. **Application.** We formalize the mapping of the framework to multi-view LLM reasoning (Section 13).
6. **Experiments.** We present proof-of-concept experiments on synthetic benchmarks (Section 16).

1.4 Why the Paper Is Self-Contained

Sections 2–7 build all necessary prerequisites from scratch: linear algebra, graph Laplacians, supra-Laplacians, simplicial complexes, chain spaces, boundary operators, and single-layer Hodge Laplacians. No algebraic topology background beyond what is defined in these sections is assumed. A reader with undergraduate linear algebra and basic familiarity with graphs should be able to follow every proof by reading sequentially.

1.5 Roadmap for the Reader

Sections 2–7 lay the mathematical foundations. Section 8 constructs the alignment and coupling infrastructure; Section 9 then defines the Supra-Hodge Laplacian L_p itself. Section 10 establishes fundamental properties and the main theorem. Section 11 develops spectral interpretation. Section 12 treats dynamics and perturbations. Section 13 maps the theory to LLM reasoning. Section 14 presents algorithms. Section 15 collects worked examples. Section 16 reports experiments. Sections 17 and 18 discuss limitations and conclude. Appendices collect proofs, notation, algorithmic details, and expanded examples.

Figure 1 illustrates the progressive construction of the Supra-Hodge system, showing how higher-order structure and multi-layer coupling are introduced step-by-step rather than as an abstract combination.

2 Linear Algebra Foundations for Spectral Methods

This section establishes the linear-algebraic language used throughout the paper. A reader comfortable with eigendecompositions and quadratic forms may treat this section as a reference.

2.1 Vectors as Signals

Let $V = \{v_1, \dots, v_n\}$ be a finite set of n entities. A *signal* on V is a function $x: V \rightarrow \mathbb{R}$, represented as a column vector $x \in \mathbb{R}^n$ with $x_i = x(v_i)$. We equip $\mathbb{R}^n$ with the standard inner product $\langle x, y \rangle = x^\top y$ and induced norm $\|x\| = \sqrt{x^\top x}$. Signals are the basic objects on which all operators act.

2.2 Matrices as Operators

A matrix $A \in \mathbb{R}^{n \times n}$ acts on signals by $x \mapsto Ax$. The (i, j) entry of A records how much the j -th coordinate of the input influences the i -th coordinate of the output. Composition of operators corresponds to matrix multiplication.

2.3 Symmetry and Positive Semidefiniteness

A matrix A is *symmetric* if $A = A^\top$. Symmetric matrices satisfy $x^\top Ay = y^\top Ax$ for all x, y , which is the self-adjoint property. All operators constructed in this paper will be symmetric.

Definition 2.1 (Positive Semidefiniteness). A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is *positive semidefinite* (PSD), written $A \succeq 0$, if

$$x^\top Ax \geq 0 \quad \text{for all } x \in \mathbb{R}^n.$$

It is *positive definite* if $x^\top Ax > 0$ for all $x \neq 0$.

Every PSD matrix can be written as $A = M^\top M$ for some (not necessarily square) matrix M . This factored form is the key reason all Hodge Laplacians are PSD: they are defined precisely as sums of such products.

2.4 Quadratic Forms and Energy

For a symmetric matrix A , the function $\mathcal{E}_A(x) = x^\top Ax$ is the *quadratic form* of A . It assigns a scalar “energy” to each signal x . When $A \succeq 0$, this energy is nonnegative, and signals with zero energy lie in the *nullspace* $\ker(A) = \{x : Ax = 0\}$. The nullspace is a linear subspace encoding the operator’s equilibrium configurations.

2.5 Energy as Smoothness and Structure

A quadratic form of the type $x^\top Ax$ should not be viewed as an abstract algebraic object, but as a measure of structural compatibility of a signal x with respect to an operator A .

In the context of graph-based operators, this quantity admits a concrete interpretation. Let L denote a graph Laplacian. Then one can expand:

$$x^\top Lx = \sum_{(u,v) \in E} w_{uv}(x_u - x_v)^2.$$

This expression reveals that the quadratic form penalizes variation across edges: it is small when adjacent nodes take similar values, and large when they differ significantly. Thus, $x^\top Lx$ measures the *roughness* of the signal x over the graph.

Convention on the energy sum. If one sums over unordered edges $\{u, v\} \in E$ once, then $x^\top Lx = \sum_{\{u,v\} \in E} w_{uv}(x_u - x_v)^2$. If one sums over ordered pairs via the adjacency matrix, the symmetric form $x^\top Lx = \frac{1}{2} \sum_{u,v} A_{uv}(x_u - x_v)^2$ from Proposition 3.3 is the same quantity. We use whichever is notationally convenient.

Eigenvectors of L arise as minimizers of this energy under orthogonality constraints. The smallest eigenvalues correspond to the smoothest signals. In particular, the second eigenvector (the Fiedler vector) identifies the weakest point of connectivity in the graph, separating it into coherent regions.

This variational perspective will generalize directly to the higher-order and multi-layer operators constructed later in the paper.

2.6 Eigenvalues, Eigenvectors, and Orthogonality

Definition 2.2 (Eigendecomposition). A nonzero vector $u \in \mathbb{R}^n$ and scalar $\lambda \in \mathbb{R}$ satisfying

$$Au = \lambda u$$

are an *eigenvector* and *eigenvalue* of A , respectively.

The *spectral theorem* guarantees that any symmetric A has a complete orthonormal eigenvector basis $u_1, \dots, u_n$ with real eigenvalues $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$. Writing $A = U\Lambda U^\top$ (with $U = [u_1 \cdots u_n]$ and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$) makes the energy transparent: $x^\top Ax = \sum_i \lambda_i (u_i^\top x)^2$, a nonneg combination of projected energies along each eigenmode.

2.7 Rayleigh Quotients and Variational Thinking

Definition 2.3 (Rayleigh Quotient). For symmetric A and nonzero $x \in \mathbb{R}^n$, the *Rayleigh quotient* is

$$R_A(x) = \frac{x^\top Ax}{x^\top x}.$$

The variational principle states $\lambda_1 = \min_{x \neq 0} R_A(x)$, achieved at $x = u_1$. More generally, $\lambda_j = \min\{R_A(x) : x \perp u_1, \dots, u_{j-1}, x \neq 0\}$. Eigenvectors are therefore the directions of extremal energy, and eigenvalues are the corresponding energy levels.

Example 2.4 (Two-by-two spectral example). Let

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

The characteristic polynomial is $\lambda^2 - 4\lambda + 3 = 0$, giving $\lambda_1 = 1$ and $\lambda_2 = 3$ with eigenvectors $u_1 = \frac{1}{\sqrt{2}}[1, 1]^\top$ and $u_2 = \frac{1}{\sqrt{2}}[1, -1]^\top$. The in-phase direction (u_1) incurs energy 1 while the out-of-phase direction (u_2) incurs energy 3: the operator rewards agreement between coordinates and penalizes disagreement. This 2×2 template is the algebraic skeleton of every operator in this paper.

3 Graphs and the Classical Graph Laplacian

This section builds the first and most fundamental spectral operator on relational data: the graph Laplacian. All operators introduced later reduce to this one in the appropriate special case.

3.1 Weighted Graphs and Adjacency Structure

Definition 3.1 (Weighted Graph). A *weighted undirected graph* is a triple $G = (V, E, w)$ where $V = \{v_1, \dots, v_n\}$ is the vertex set, $E \subseteq \binom{V}{2}$ is the edge set, and $w: E \rightarrow \mathbb{R}_{>0}$ assigns positive weights. The *adjacency matrix* $A \in \mathbb{R}^{n \times n}$ has $A_{uv} = w(u, v)$ if $\{u, v\} \in E$ and $A_{uv} = 0$ otherwise.

3.2 Degree Matrices and the Operator $L = D - A$

Definition 3.2 (Degree Matrix and Graph Laplacian). The *degree matrix* $D = \text{diag}(d_1, \dots, d_n)$ has

$$d_u = \sum_v A_{uv}.$$

The *graph Laplacian* is $L = D - A$.

Every row of L sums to zero ($L\mathbf{1} = 0$), reflecting that a constant signal has zero energy on any graph. The matrix L is symmetric because A is symmetric.

3.3 Interpretation of the Graph Laplacian

The graph Laplacian is not merely an algebraic construction, but a discrete differential operator encoding how signals vary across a network.

The identity

$$x^\top Lx = \sum_{(u,v) \in E} w_{uv}(x_u - x_v)^2$$

shows that the Laplacian measures disagreement between neighboring nodes. A signal that is constant across connected components has zero energy and lies in the kernel of L .

The eigenvalues of L quantify connectivity. In particular:

- The multiplicity of the eigenvalue 0 equals the number of connected components.
- The second-smallest eigenvalue (the Fiedler value) measures how well-connected the graph is.
- The corresponding eigenvector identifies a minimal-energy partition of the graph.

This interpretation will serve as the foundation for understanding higher-order Laplacians and their spectral behavior.

3.4 Smoothness Energy on Graphs

Proposition 3.3 (Graph Laplacian Energy Identity). *For any signal $x \in \mathbb{R}^n$,*

$$x^\top Lx = \frac{1}{2} \sum_{u,v} A_{uv}(x_u - x_v)^2.$$

Proof. We expand $x^\top (D - A)x = x^\top Dx - x^\top Ax = \sum_u d_u x_u^2 - \sum_{u,v} A_{uv} x_u x_v$. Using $d_u = \sum_v A_{uv}$ in the first term:

$$x^\top Lx = \sum_{u,v} A_{uv} x_u^2 - \sum_{u,v} A_{uv} x_u x_v = \frac{1}{2} \sum_{u,v} A_{uv} (x_u^2 - 2x_u x_v + x_v^2) = \frac{1}{2} \sum_{u,v} A_{uv} (x_u - x_v)^2. \quad \square$$

The energy $x^\top Lx$ is a weighted sum of squared differences across all edges. A signal with low energy varies slowly across highly-weighted edges; this is the formal meaning of *graph smoothness*. Since every term is nonnegative, $L \succeq 0$, and the eigenvalues satisfy

$$0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n.$$

3.5 Kernel, Connectedness, and Constant Signals

Proposition 3.4 (Kernel of the Graph Laplacian). $x \in \ker(L)$ if and only if $x_u = x_v$ for every edge $\{u, v\} \in E$. Consequently: (1) constant signals always lie in $\ker(L)$; (2) for a connected graph, $\ker(L) = \text{span}(\mathbf{1})$ and $\lambda_1 = 0$ has multiplicity one; (3) for a graph with c connected components, $\dim \ker(L) = c$.

Proof. By Proposition 3.3, $x^\top Lx = 0$ iff $A_{uv}(x_u - x_v)^2 = 0$ for all u, v , i.e., $x_u = x_v$ on every edge. Connectedness forces every such x to be constant. The component count follows by applying the same argument within each component. $\square$

Constant signals represent perfect global consensus: every vertex holds the same value. The nullity of L counts how many independent consensus modes exist—one per connected component.

3.6 The Second Eigenvalue and the Fiedler Vector

Definition 3.5 (Fiedler Value and Vector). For a connected graph, $\lambda_2 > 0$ is the *Fiedler value* (algebraic connectivity). The corresponding unit eigenvector $u_2 \perp \mathbf{1}$ is the *Fiedler vector*.

The Fiedler vector partitions vertices by sign: vertices with $u_2(v) > 0$ form one cluster, those with $u_2(v) < 0$ form another [1]. A large λ_2 indicates high algebraic connectivity (the graph is hard to cut); a small λ_2 flags a near-bottleneck. The Fiedler theory generalizes directly to the Supra-Hodge setting in Section 11.

3.7 Worked Graph Example

Example 3.6 (Four-node weighted graph). Let G have $V = \{1, 2, 3, 4\}$ and edges $\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}, \{1, 3\}$, all with unit weight. Then

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}, \quad D = \text{diag}(3, 2, 3, 2), \quad L = \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix}.$$

The constant signal $x = \mathbf{1}$ satisfies $Lx = 0$. The signal $y = [1, -1, 1, -1]^\top$ has energy $y^\top Ly = 8$, reflecting large differences across all edges. The Fiedler vector separates $\{1, 4\}$ from $\{2, 3\}$, marking the minimal bottleneck. We reuse this example in Section 15.

4 Multilayer Graphs and Supra-Laplacians

This section extends the graph Laplacian to systems with multiple relation types over the same entity set. The resulting supra-Laplacian is the direct predecessor of the Supra-Hodge Laplacian, and its energy identity is the template for the main variational result of the paper.

4.1 Multiple Relation Types on One Entity Set

Let V be a fixed entity set with $|V| = n$. We consider k weighted graphs $\mathcal{G} = \{G^{(1)}, \dots, G^{(k)}\}$, each on vertex set V :

$$\mathcal{G} = \{G^{(1)}, G^{(2)}, \dots, G^{(k)}\}.$$

Each $G^{(i)} = (V, E^{(i)}, w^{(i)})$ encodes a distinct relation type. In a reasoning system, these might be semantic similarity, evidential support, causal dependency, and temporal precedence.

4.2 Layerwise Graph Laplacians

Each layer $G^{(i)}$ has its own adjacency matrix $A^{(i)}$, degree matrix $D^{(i)}$, and graph Laplacian $L^{(i)} = D^{(i)} - A^{(i)}$. By Proposition 3.3, $L^{(i)} \succeq 0$ and $\ker(L^{(i)})$ captures the consensus structure within layer i .

4.3 Stacked Signals and Block Operators

To analyze all layers jointly, we stack layer signals into a single vector $v = (v^{(1)}, \dots, v^{(k)}) \in \mathbb{R}^{kn}$. Inter-layer coupling is encoded by weights $\omega_{ij} \geq 0$: a positive ω_{ij} attracts the signals of layers i and j toward agreement.

4.4 The Supra-Laplacian and Its Energy

Definition 4.1 (Supra-Laplacian). The *supra-Laplacian* is the $kn \times kn$ block-symmetric matrix

$$L_{\text{sup}} = \begin{bmatrix} L^{(1)} + \sum_{j \neq 1} \omega_{1j} I & -\omega_{12} I & \cdots & -\omega_{1k} I \\ -\omega_{21} I & L^{(2)} + \sum_{j \neq 2} \omega_{2j} I & \cdots & -\omega_{2k} I \\ \vdots & \vdots & \ddots & \vdots \\ -\omega_{k1} I & -\omega_{k2} I & \cdots & L^{(k)} + \sum_{j \neq k} \omega_{kj} I \end{bmatrix}. \quad (1)$$

Proposition 4.2 (Supra-Laplacian Energy Decomposition). *For any stacked signal $v = (v^{(1)}, \dots, v^{(k)})$,*

$$v^\top L_{\text{sup}} v = \sum_{i=1}^k v^{(i)\top} L^{(i)} v^{(i)} + \sum_{i < j} \omega_{ij} \|v^{(i)} - v^{(j)}\|_2^2.$$

Proof. Each diagonal block $L^{(i)} + \sum_{j \neq i} \omega_{ij} I$ contributes $v^{(i)\top} L^{(i)} v^{(i)} + \sum_{j \neq i} \omega_{ij} \|v^{(i)}\|^2$ to $v^\top L_{\text{sup}} v$. Each off-diagonal block $-\omega_{ij} I$ contributes $-2\omega_{ij} v^{(i)\top} v^{(j)}$. Collecting all pairs (i, j) with $i < j$:

$$\sum_{i < j} \omega_{ij} (\|v^{(i)}\|^2 - 2v^{(i)\top} v^{(j)} + \|v^{(j)}\|^2) = \sum_{i < j} \omega_{ij} \|v^{(i)} - v^{(j)}\|^2. \quad \square$$

The energy decomposes into within-layer graph smoothness and cross-layer disagreement. This structure is the exact prototype for the Supra-Hodge energy expansion proved in Section 10.

4.5 Consensus and Disagreement Across Layers

A stacked signal v achieves zero supra-Laplacian energy iff $L^{(i)} v^{(i)} = 0$ for all i (within-layer graph consensus) and $v^{(i)} = v^{(j)}$ whenever $\omega_{ij} > 0$ (cross-layer consensus). The supra-Laplacian therefore enforces two simultaneous smoothness conditions.

4.6 Worked Two-Layer Example

Example 4.3 (Two-layer supra-Laplacian). Let $n = 3$, $k = 2$, both layers equal to the path graph 1–2–3 with $L^{(1)} = L^{(2)} = L_{\text{path}}$, and $\omega_{12} = 1$. Then

$$L_{\text{sup}} = \begin{bmatrix} L_{\text{path}} + I & -I \\ -I & L_{\text{path}} + I \end{bmatrix}.$$

The symmetric signal $v = (\mathbf{1}, \mathbf{1})$ has zero energy (constant in each layer, equal across layers). The antisymmetric signal $v = (\mathbf{1}, -\mathbf{1})$ incurs cross-layer energy $\omega_{12} \|\mathbf{21}\|^2 = 12$.

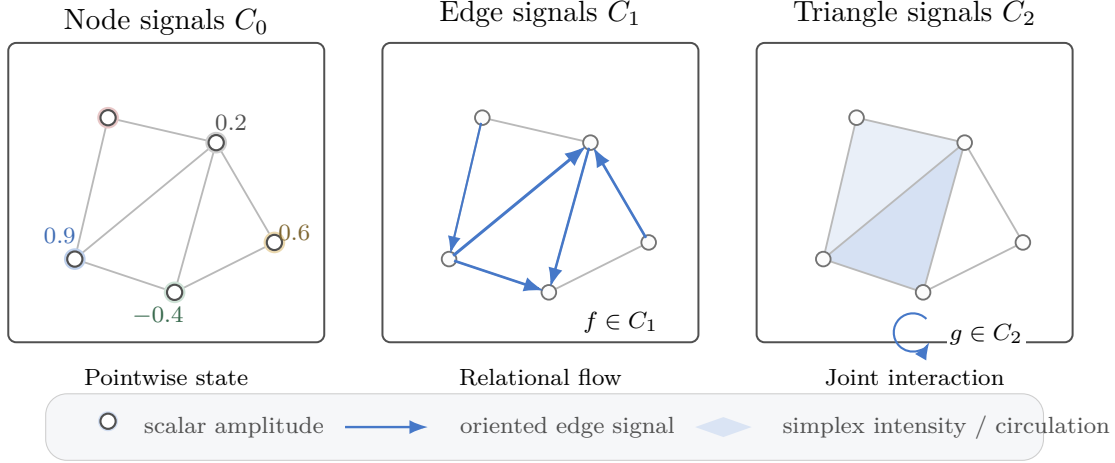


Figure 2: Signals on simplices of different order over the same underlying simplicial complex: node signals in C_0 , oriented edge signals in C_1 , and triangle-level signals in C_2 . The display treats the three signal types as parallel views of the same base higher-order structure.

4.7 Why Pairwise Multilayer Models Still Fall Short

Even with k coupled layers, the supra-Laplacian encodes only pairwise edges within each layer. It cannot distinguish a filled triangle (a 2-simplex) from a hollow cycle of three edges. If three entities jointly support a conclusion in a way that no pair does independently, the supra-Laplacian assigns the same energy to both configurations. Correcting this requires Hodge theory.

5 Simplicial Complexes and Higher-Order Relational Structure

This section introduces simplicial complexes: the combinatorial structures that generalize graphs to encode interactions involving arbitrarily many entities simultaneously.

5.1 From Graphs to Simplicial Complexes: A Concrete Example

A graph encodes pairwise relationships through edges. However, many systems exhibit interactions that involve more than two entities simultaneously.

Consider three vertices u, v, w . A graph may contain all three edges (u, v) , (v, w) , and (u, w) , forming a triangle. However, this does not necessarily imply that the three entities form a coherent higher-order unit.

A simplicial complex distinguishes between these cases:

- The presence of all three edges represents pairwise compatibility.
- The inclusion of the 2-simplex (u, v, w) asserts a genuine three-way interaction.

This distinction is critical: higher-order structure cannot be inferred from pairwise relations alone. Simplicial complexes provide the appropriate mathematical language for encoding such multi-way interactions.

Figure 2 visualizes signals on p -simplices, distinguishing node values, edge flows, and triangle-level circulation.

5.2 From Pairwise Edges to Simplices

Definition 5.1 (Simplex and Simplicial Complex). A p -simplex is a set $\sigma = \{v_0, v_1, \dots, v_p\} \subseteq V$ of $p + 1$ elements. A *simplicial complex* K on V is a collection of simplices satisfying the *closure property*:

$$\sigma \in K, \quad \tau \subseteq \sigma \implies \tau \in K.$$

The dimension of a simplex is $\dim(\sigma) = |\sigma| - 1$. Thus 0-simplices are vertices, 1-simplices are (undirected) edges, 2-simplices are filled triangles, and 3-simplices are filled tetrahedra.

5.3 Faces, Closure, and Dimension

Each subset of a simplex is a *face*. The closure property says: if a higher-order interaction is present, all its sub-interactions are present too. This is the natural consistency requirement for relational data. We write K_p for the set of all p -simplices of K .

5.4 Hollow Triangles Versus Filled Triangles

Example 5.2 (Hollow vs. filled triangle). Let $V = \{1, 2, 3\}$. The *hollow complex* $K_{\text{hollow}} = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}\}$ contains all three edges but not the triangle as a 2-simplex. The *filled complex* $K_{\text{filled}} = K_{\text{hollow}} \cup \{\{1, 2, 3\}\}$ adds the triangle. These two complexes have identical pairwise adjacency structure. Yet they have different topology: K_{hollow} contains a non-trivial 1-cycle (a “hole”), while K_{filled} does not. Graph Laplacian methods cannot distinguish them; Hodge methods detect the difference precisely via harmonic structure.

5.5 Why Higher-Order Structure Changes the Model Class

Adding the 2-simplex $\{1, 2, 3\}$ changes the topology of the complex and changes the spectrum of the associated Hodge Laplacian. In the hollow complex, the edge signal $f = [1, 1, 1]^\top$ (uniform flow around the cycle) is harmonic at $p = 1$ (zero H_1 -energy). In the filled complex, this signal has nonzero upper Laplacian energy: the triangle “detects” that the flow winds around it, and the harmonic cycle is destroyed. No pairwise operator can see this difference.

5.6 Examples from Reasoning and Relational Systems

In an evidential layer of a reasoning trace, three propositions $\{P_1, P_2, P_3\}$ may jointly support a conclusion even though no pair does alone. This is a 2-simplex in the evidential complex. Its presence eliminates the harmonic 1-cycle formed by the three pairwise support edges, flagging the triplet as genuinely consistent. Pairwise models would treat a three-way evidence coalition identically to three isolated pairwise connections.

6 Chain Spaces, Orientation, and Boundary Operators

This section develops the algebraic machinery for computing with simplicial complexes: chain spaces that coordinatize signals on simplices of each dimension, and boundary operators that map between consecutive dimensions.

6.1 p -Simplices and p -Chains

Definition 6.1 (Chain Space). The p -chain space of K is

$$C_p(K) = \mathbb{R}^{|K_p|},$$

with one coordinate per p -simplex of K . A p -chain is an element of $C_p(K)$.

Signals on vertices live in $C_0(K) = \mathbb{R}^n$; signals on edges live in $C_1(K)$; signals on triangles live in $C_2(K)$. The chain space generalizes the notion of a vertex signal to signals of every order.

6.2 Basis Elements and Coordinate Representations

We choose an orientation for each p -simplex. An oriented simplex is written $[v_0, v_1, \dots, v_p]$, and reversing any adjacent pair negates the element: $[v_0, \dots, v_i, v_{i+1}, \dots, v_p] = -[v_0, \dots, v_{i+1}, v_i, \dots, v_p]$. The set of consistently oriented p -simplices forms a basis for $C_p(K)$.

6.3 Orientation and Sign

Orientation is a bookkeeping device that makes boundary operators well-defined. For an edge $[u, v]$, it assigns a direction (arrow from u to v). For a triangle $[u, v, w]$, it assigns a cyclic ordering. Changing orientation changes the sign of the basis element, ensuring that boundary operators are orientation-consistent.

6.4 Boundary Operators

Definition 6.2 (Boundary Operator). The *boundary operator* $\partial_p: C_p(K) \rightarrow C_{p-1}(K)$ is defined on basis elements by

$$\partial_p[v_0, v_1, \dots, v_p] = \sum_{i=0}^p (-1)^i [v_0, \dots, \hat{v}_i, \dots, v_p],$$

where $\hat{v}_i$ denotes omission of v_i , and extended by linearity.

6.5 Example: Boundary of a Triangle

For a 2-simplex $[v_0, v_1, v_2]$, one has $\partial_2[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1]$, the oriented edge cycle around the triangle; applying ∂_1 again gives 0, i.e. $\partial_1 \partial_2 = 0$ (matrix form: $B_1 B_2 = 0$). Example 6.4 computes the incidence matrices explicitly.

The boundary operator maps each oriented p -simplex to the signed sum of its $(p-1)$ -dimensional faces, with signs determined by the orientation. Intuitively, the boundary of an oriented edge $[u, v]$ is the signed difference $v - u$; the boundary of an oriented triangle $[u, v, w]$ is the signed cycle $[v, w] - [u, w] + [u, v]$.

6.6 Incidence Matrices

We represent ∂_p as the matrix $B_p \in \mathbb{R}^{|K_{p-1}| \times |K_p|}$: the column for oriented p -simplex σ has $+1$ in row τ if τ appears in $\partial_p \sigma$ with positive orientation, -1 if with negative orientation, and 0 otherwise.

6.7 Why Boundary of Boundary Is Zero

Proposition 6.3 (Boundary of Boundary Identity). *For any p ,*

$$\partial_p \partial_{p+1} = 0, \quad B_p B_{p+1} = 0.$$

Proof. For an oriented $(p+1)$ -simplex $[v_0, \dots, v_{p+1}]$, compute $\partial_p \partial_{p+1}[v_0, \dots, v_{p+1}]$:

$$\partial_p \partial_{p+1}[v_0, \dots, v_{p+1}] = \sum_{i=0}^{p+1} (-1)^i \partial_p[v_0, \dots, \hat{v}_i, \dots, v_{p+1}].$$

Each p -face $[v_0, \dots, \hat{v}_i, \dots, \hat{v}_j, \dots, v_{p+1}]$ (with both v_i and v_j omitted) appears twice: once as the j -th term of $\partial_p[v_0, \dots, \hat{v}_i, \dots]$ with sign $(-1)^i(-1)^{j-1}$, and once as the i -th term of $\partial_p[v_0, \dots, \hat{v}_j, \dots]$ with sign $(-1)^j(-1)^i$. These two contributions are equal in magnitude and opposite in sign, so they cancel. Hence $\partial_p \partial_{p+1} = 0$. The matrix identity $B_p B_{p+1} = 0$ is the coordinate form. $\square$

This identity is the algebraic cornerstone of homological algebra. It implies $\text{im}(B_{p+1}) \subseteq \ker(B_p)$, enabling the Hodge decomposition.

6.8 Worked Simplex-Boundary Example

Example 6.4 (Oriented triangle boundary). Let K have vertices $\{1, 2, 3\}$, oriented edges $[1, 2], [1, 3], [2, 3]$, and oriented triangle $[1, 2, 3]$. Then

$$\partial_2[1, 2, 3] = [2, 3] - [1, 3] + [1, 2].$$

With edges listed in order $[1, 2], [1, 3], [2, 3]$ as the basis of C_1 :

$$B_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \quad B_1 = \begin{bmatrix} -1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Direct computation confirms $B_1 B_2 = [1 - 1, -1 + 1, 1 - 1]^\top = 0$, illustrating Proposition 6.3.

7 Hodge Laplacians on Simplicial Complexes

We now define the combinatorial Hodge Laplacian, the higher-order spectral operator on a single complex that the Supra-Hodge Laplacian will generalize.

7.1 Definition of the Combinatorial Hodge Laplacian

Definition 7.1 (Hodge Laplacian). The p -th combinatorial Hodge Laplacian of K is

$$H_p = B_p^\top B_p + B_{p+1} B_{p+1}^\top,$$

where we set $B_0 \equiv 0$ by convention. The term $B_p^\top B_p$ is the *lower Laplacian* and $B_{p+1} B_{p+1}^\top$ is the *upper Laplacian*.

7.2 Interpretation of Hodge Decomposition

By the standard combinatorial Hodge decomposition for finite simplicial complexes [4–6], chain space splits into three orthogonal components:

$$C_p = \text{im}(B_p^\top) \oplus \ker(H_p) \oplus \text{im}(B_{p+1}).$$

These components admit intuitive interpretations:

- **Gradient component:** signals derived from lower-order structure.
- **Curl component:** signals corresponding to local cycles or inconsistencies.
- **Harmonic component:** globally consistent structures that are neither gradients nor curls.

In particular, for $p = 1$, the curl term captures inconsistencies on triangles and higher simplices. This enables the detection of cyclic contradictions that are invisible to graph-based methods.

The lower Laplacian measures how much a p -chain signal diverges at $(p - 1)$ -boundaries; the upper Laplacian measures how much it has nonzero curl around $(p + 1)$ -simplices.

Proposition 7.2 (Hodge Laplacians Are Symmetric PSD). *H_p is symmetric and positive semidefinite for every p .*

Proof. Both $B_p^\top B_p$ and $B_{p+1} B_{p+1}^\top$ are of the form $M^\top M$ (with $M = B_p$ and $M = B_{p+1}^\top$ respectively), hence symmetric PSD. A sum of symmetric PSD matrices is symmetric PSD. $\square$

7.3 Recovery of the Graph Laplacian at $p = 0$

Proposition 7.3 (Hodge-0 Equals Graph Laplacian). *$H_0 = B_1 B_1^\top$ equals the graph Laplacian L of the 1-skeleton of K .*

Proof. For oriented edge $[u, v]$ with $u < v$ (so that B_1 has -1 in row u and $+1$ in row v), the (a, b) -entry of $B_1 B_1^\top$ is $\sum_e (B_1)_{ae} (B_1)_{be}$. For $a = b$: $(B_1 B_1^\top)_{aa} = \sum_e (B_1)_{ae}^2 = |\{e : a \in e\}| = d_a$. For $a \neq b$: $(B_1 B_1^\top)_{ab} = -A_{ab}$ (nonzero only when $\{a, b\} \in E$). Hence $B_1 B_1^\top = D - A = L$. $\square$

This proposition confirms that Hodge theory is a genuine extension of graph spectral theory: H_0 is the original graph Laplacian.

7.4 The $p = 1$ Case and Cycle-Level Inconsistency

For $p = 1$, a signal $f \in C_1(K)$ (a flow or relation strength on edges) has

$$f^\top H_1 f = \|B_1 f\|^2 + \|B_2^\top f\|^2.$$

The term $\|B_1 f\|^2$ measures vertex-level divergence of f (how much flow is not conserved at each vertex). The term $\|B_2^\top f\|^2$ measures triangle-level curl (how much f winds around each filled triangle). Both terms must vanish for f to be harmonic.

7.5 Lower and Upper Adjacency

Two p -simplices are *lower-adjacent* if they share a $(p - 1)$ -face; they are *upper-adjacent* if they are co-faces of a common $(p + 1)$ -simplex. Lower adjacency governs the lower Laplacian $B_p^\top B_p$; upper adjacency governs the upper Laplacian $B_{p+1} B_{p+1}^\top$. Graph Laplacians correspond to the $p = 0$ case where only lower adjacency (vertex-to-edge incidence) exists.

7.6 Gradient, Curl, and Harmonic Components

The Hodge decomposition theorem states that for any $f \in C_p(K)$,

$$f = f_{\text{grad}} + f_{\text{harm}} + f_{\text{curl}},$$

where $f_{\text{grad}} \in \text{im}(B_p^\top)$, $f_{\text{curl}} \in \text{im}(B_{p+1})$, and $f_{\text{harm}} \in \ker(H_p)$. These subspaces are mutually orthogonal. The gradient component is the conservative part (derivable from a vertex potential); the curl component is the exact part (boundary of a higher-order chain); the harmonic component is the topologically persistent part (belonging to genuine higher-order cycles).

Remark on the decomposition. For finite simplicial complexes equipped with the standard inner product on chain spaces, the combinatorial Hodge decomposition is an orthogonal decomposition. Thus the gradient, harmonic, and curl components are not merely formal labels, but mutually orthogonal subspaces with distinct spectral roles.

7.7 Hodge Decomposition Intuition

On edge signals ($p = 1$), the gradient component represents potential-driven flow (water flowing downhill from a potential), the curl component represents circulation induced by filled triangles (water flowing around a filled basin), and the harmonic component represents persistent circulation that is not caused by any local source or triangle fill (water in a donut). The Hodge Laplacian measures and separates all three simultaneously.

7.8 Worked Hodge Example

Example 7.4 (Filled-triangle Hodge computation). Let $K = K_{\text{filled}}$ from Example 5.2, with B_1 and B_2 as in Example 6.4. Then

$$B_1^\top B_1 = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}, \quad B_2 B_2^\top = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix},$$

giving

$$H_1 = B_1^\top B_1 + B_2 B_2^\top = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Thus $H_1 = 3I_3$ for the filled triangle, and $\ker(H_1) = \{0\}$: there are no harmonic 1-cycles. For the hollow complex ($B_2 = 0$), $H_1 = B_1^\top B_1$ has $\ker(H_1) = \text{span}([1, 1, 1]^\top / \sqrt{3})$ —the uniform circulation around the empty triangle is a harmonic cycle.

8 From Single Higher-Order Systems to Coupled Higher-Order Systems

A single simplicial complex captures one relational view. Coupling several such views requires a common ambient p -chain space so that layerwise signals and operators can be compared and coupled without conflating distinct per-view topologies.

The purpose of this section is to construct the aligned ambient chain space and embedding machinery needed to couple multiple higher-order views. Once that infrastructure is in place, the Supra-Hodge operator itself will be defined formally in Section 9.

8.1 Why Alignment Is Necessary

Each layer in a multi-view system defines its own simplicial complex and corresponding chain space. Without a common coordinate system, these spaces cannot be meaningfully coupled.

A naive approach would attempt to compare signals across layers directly. However, if the underlying simplices differ, such comparisons are ill-defined: one layer may contain a simplex that does not exist in another.

To resolve this, we introduce an aligned ambient chain space. Each layer is embedded into this shared space by zero-padding missing simplices. This construction enables the definition of a block operator that couples corresponding structures across layers while preserving the internal topology of each layer.

This alignment step is the key technical device that allows higher-order operators to be extended to the multi-layer setting.

8.2 Why a Single Simplicial Complex Is Not Enough

A reasoning agent maintains semantic clusters, evidential support relations, and temporal precedences over the same entities. Each view has its own higher-order structure (semantic triplets, evidence triplets, temporal chains). Merging all views into a single complex conflates semantically unrelated triangles and destroys the per-view interpretation of harmonic cycles. We need a family of complexes and a principled coupling.

8.3 Families of Complexes Over a Shared Entity Set

Definition 8.1 (Family of Simplicial Complexes). Let V be a shared vertex set. A *family of simplicial complexes* is

$$\mathcal{K} = \{K^{(1)}, K^{(2)}, \dots, K^{(k)}\},$$

where each $K^{(i)}$ is a simplicial complex on V .

Each $K^{(i)}$ has its own p -simplices $K_p^{(i)}$ and chain space $C_p^{(i)} = \mathbb{R}^{|K_p^{(i)}|}$.

8.4 Alignment of Simplex Coordinates Across Views

To couple the chain spaces of different layers, we need a shared coordinate system.

Definition 8.2 (Aligned Reference Set). An *aligned reference set* Σ_p is a set of admissible oriented p -simplices such that $K_p^{(i)} \subseteq \Sigma_p$ for all i . The ambient dimension is $N_p = |\Sigma_p|$. The *common ambient chain space* is

$$\tilde{C}_p \cong \mathbb{R}^{N_p}.$$

The simplest choice is $\Sigma_p = \bigcup_i K_p^{(i)}$ (union over all layers), with $N_p = |\bigcup_i K_p^{(i)}|$.

8.5 Explicit Embedding into the Ambient Chain Space

To make the alignment construction fully explicit, we define for each layer i an embedding map

$$E_p^{(i)} : C_p(K^{(i)}) \longrightarrow \tilde{C}_p \cong \mathbb{R}^{N_p},$$

where $N_p = |\Sigma_p|$. The map $E_p^{(i)}$ sends each basis simplex of $C_p(K^{(i)})$ to its corresponding coordinate in the ambient basis indexed by Σ_p , and assigns zero to all coordinates corresponding to simplices absent from layer i .

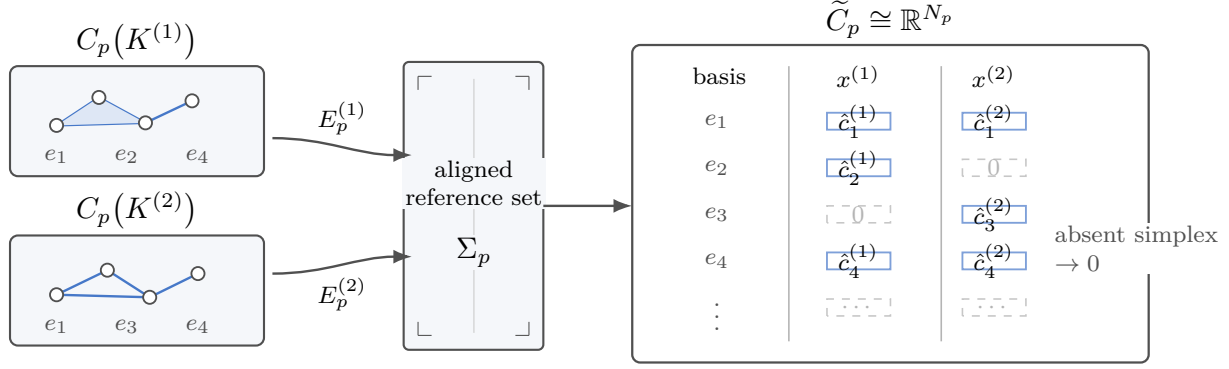


Figure 3: Alignment of layerwise chain spaces into a shared ambient basis. Each layer is embedded by a coordinate-alignment map $E_p^{(i)}$ into the common ambient chain space $\tilde{C}_p$, with missing simplices represented by zero-padding so that cross-layer coupling is defined on matched coordinates.

Using this embedding, any layerwise p -chain signal $x^{(i)} \in C_p(K^{(i)})$ is represented in the common ambient space by $E_p^{(i)}x^{(i)} \in \tilde{C}_p$. After padding incidence matrices to the reference set Σ_p (equivalently, conjugating the native Hodge matrices by $E_p^{(i)}$), every layerwise Hodge Laplacian becomes an $N_p \times N_p$ operator on $\tilde{C}_p$. We denote this embedded matrix by $H_p^{(i)}$ throughout the block definition (2) and the remainder of the paper.

This makes the coupling mathematically precise: the Supra-Hodge operator does not compare signals in unrelated spaces, but in a shared ambient chain space obtained by explicit embeddings.

The embedding of layerwise chain spaces into a common ambient space is illustrated in Figure 3, where zero-padding and coordinate alignment are made explicit.

Notational convention. From this point onward, we write $H_p^{(i)}$ for the ambient-space operator acting on $\tilde{C}_p$ obtained from the embedding construction above. Thus all layerwise Hodge operators appearing in the definition of L_p , in the main theorem, and in subsequent reductions are understood as operators on the common aligned chain space.

8.6 Zero-Padding and a Common Ambient Chain Space

Each layer i has its own chain space $C_p^{(i)}$ of dimension $|K_p^{(i)}| \leq N_p$. We embed it into $\tilde{C}_p$ by *zero-padding*: the boundary matrix $B_p^{(i)}$ is extended to $\mathbb{R}^{N_{p-1} \times N_p}$ by inserting zero columns for simplices in $\Sigma_p \setminus K_p^{(i)}$. The Hodge Laplacian $H_p^{(i)} = B_p^{(i)\top} B_p^{(i)} + B_{p+1}^{(i)} B_{p+1}^{(i)\top}$ then acts on the ambient space $\tilde{C}_p \cong \mathbb{R}^{N_p}$.

Example 8.3 (Alignment and zero-padding). Let $k = 2$, $V = \{1, 2, 3\}$, with $K_1^{(1)} = \{[1, 2], [1, 3]\}$ and $K_1^{(2)} = \{[1, 2], [2, 3]\}$. Choose $\Sigma_1 = \{[1, 2], [1, 3], [2, 3]\}$, so $N_1 = 3$. The aligned incidence matrix $B_1^{(1)}$ has a zero column for $[2, 3]$, and $B_1^{(2)}$ has a zero column for $[1, 3]$. The Hodge Laplacians $H_1^{(1)}$ and $H_1^{(2)}$ both live in $\mathbb{R}^{3 \times 3}$ and can be assembled into a $2 \times 3 \cdot 2 \times 3 = 6 \times 6$ block operator.

8.7 When Identity Coupling Is Natural

If all layers share the same p -simplices ($K_p^{(i)} = K_p^{(j)}$ for all i, j), then corresponding simplices are semantically paired and the natural coupling operator between layers i and j is $C^{(ij)} = \omega_{ij} I_{N_p}$.

This is the scalar coupling case studied throughout this paper.

8.8 When Identity Coupling Fails

If layers have different complex structures with no natural simplex correspondence, identity coupling may conflate semantically unrelated simplices. In this case, a learned or structured coupling matrix $C^{(ij)} \in \mathbb{R}^{N_p \times N_p}$ replaces $\omega_{ij}I_{N_p}$. This extension is discussed in Section 17; we focus on the scalar case here.

9 Definition of the Supra-Hodge Laplacian

Having fixed the alignment, embedding, and notational conventions of Section 8, we now define the Supra-Hodge Laplacian L_p , the central operator of this paper.

9.1 Layerwise Hodge Operators

Given the family $\mathcal{K}$ and aligned incidence matrices $B_p^{(i)} \in \mathbb{R}^{N_{p-1} \times N_p}$, the layerwise Hodge Laplacian is

$$H_p^{(i)} = B_p^{(i)\top} B_p^{(i)} + B_{p+1}^{(i)} B_{p+1}^{(i)\top} \in \mathbb{R}^{N_p \times N_p}.$$

By Proposition 7.2, $H_p^{(i)} \succeq 0$. The aligned construction ensures all $H_p^{(i)}$ act on the same ambient space $\tilde{C}_p$.

9.2 Stacked p -Chain Signal Spaces

The total signal space for the k -layer system at order p is

$$\mathcal{C}_p = \underbrace{\tilde{C}_p \oplus \cdots \oplus \tilde{C}_p}_k \cong \mathbb{R}^{kN_p}.$$

A stacked signal $v = (v^{(1)}, \dots, v^{(k)}) \in \mathcal{C}_p$ collects one ambient p -chain signal from each layer.

9.3 Inter-Layer Couplings and the Coupling Graph

Definition 9.1 (Coupling Graph and Coupling Operator). The *coupling graph* G_ω has vertex set $\{1, \dots, k\}$ and edge $\{i, j\}$ iff $\omega_{ij} > 0$. The *coupling operator* between layers i and j is $C^{(ij)} = \omega_{ij}I_{N_p}$.

The coupling graph encodes which layer pairs are linked. Its connectivity structure will determine the dimension of $\ker(L_p)$ through Theorem 10.7.

9.4 Block-Matrix Definition of L_p

Definition 9.2 (Supra-Hodge Laplacian). The *order- p Supra-Hodge Laplacian* is the $kN_p \times kN_p$ block-symmetric matrix

$$L_p = \begin{bmatrix} H_p^{(1)} + \sum_{j \neq 1} \omega_{1j} I & -\omega_{12} I & \cdots & -\omega_{1k} I \\ -\omega_{21} I & H_p^{(2)} + \sum_{j \neq 2} \omega_{2j} I & \cdots & -\omega_{2k} I \\ \vdots & \vdots & \ddots & \vdots \\ -\omega_{k1} I & -\omega_{k2} I & \cdots & H_p^{(k)} + \sum_{j \neq k} \omega_{kj} I \end{bmatrix}. \quad (2)$$

Supra-Hodge assembly

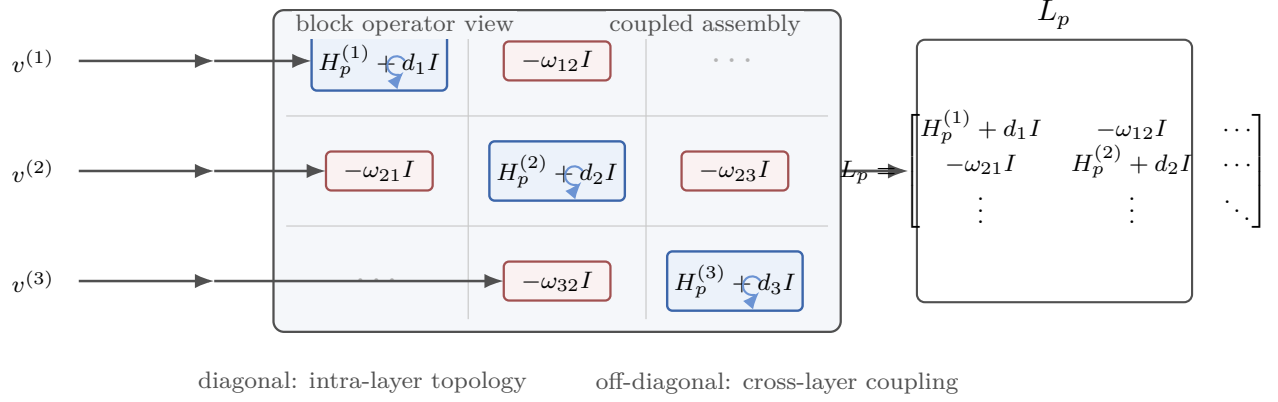


Figure 4: Block-operator view of the Supra-Hodge assembly. Diagonal blocks encode layerwise Hodge structure augmented by coupling-degree terms, while off-diagonal blocks encode inter-layer attraction. The machine-like layout preserves the matrix semantics of L_p while making the separation between intra-layer topology and cross-layer coupling visually explicit.

Each diagonal block $H_p^{(i)} + \sum_{j \neq i} \omega_{ij} I$ is the layerwise Hodge operator augmented by the total coupling weight from layer i to all other layers. Each off-diagonal block $-\omega_{ij} I$ provides the cross-layer attractive force. The structural analogy with the supra-Laplacian equation (1) is exact, with $H_p^{(i)}$ replacing $L^{(i)}$ and N_p replacing n .

The block structure of L_p is visualized in Figure 4, making explicit the separation between intra-layer Hodge structure and inter-layer coupling terms.

9.5 Equivalent Operator Form

Let

$$\mathcal{H}_p := \text{diag}(H_p^{(1)}, \dots, H_p^{(k)})$$

denote the block-diagonal operator collecting the layerwise Hodge Laplacians, and let L_{inter} denote the graph Laplacian of the coupling graph G_ω . Then the Supra-Hodge Laplacian may be written compactly as

$$L_p = \mathcal{H}_p + L_{\text{inter}} \otimes I_{N_p}.$$

This form makes the structure of the operator especially transparent: the first term captures within-layer higher-order topology, while the second captures cross-layer disagreement through a Kronecker coupling.

9.6 Aggregated Multi-Order Operator L_{SH}

When signals of multiple orders are relevant simultaneously, we aggregate:

$$L_{\text{SH}} = \text{diag}(\alpha_0 L_0, \alpha_1 L_1, \dots, \alpha_P L_P),$$

where $\alpha_p \geq 0$ are order weights and P is the maximum simplex dimension. This block-diagonal operator acts on the direct sum of all stacked chain spaces.

9.7 Interpretation Before Proof

The operator L_p charges a stacked signal v for two kinds of roughness simultaneously. The diagonal blocks charge for within-layer topological roughness: each $v^{(i)}$ is penalized for deviating from harmonicity in layer i , where harmonicity means both divergence-free at $(p-1)$ -boundaries and curl-free around $(p+1)$ -simplices. The off-diagonal blocks charge for cross-layer disagreement: $v^{(i)}$ and $v^{(j)}$ are attracted toward equality by the coupling weight ω_{ij} . The total energy is the sum of both penalties, and zero energy requires both to vanish simultaneously.

10 Fundamental Properties and Main Theorem

We establish the mathematical foundations of L_p and state and prove the main theorem.

10.1 Energy Expansion

Proposition 10.1 (Supra-Hodge Energy Expansion). *For any stacked p -chain signal $v = (v^{(1)}, \dots, v^{(k)}) \in \mathcal{C}_p$,*

$$v^\top L_p v = \sum_{i=1}^k v^{(i)\top} H_p^{(i)} v^{(i)} + \sum_{i < j} \omega_{ij} \|v^{(i)} - v^{(j)}\|_2^2.$$

Proof. Write L_p in block form as in Definition 9.2. Then

$$v^\top L_p v = \sum_{i=1}^k v^{(i)\top} \left(H_p^{(i)} + \sum_{j \neq i} \omega_{ij} I \right) v^{(i)} + \sum_{i \neq j} v^{(i)\top} (-\omega_{ij} I) v^{(j)}.$$

Separating the diagonal Hodge terms gives

$$\sum_{i=1}^k v^{(i)\top} H_p^{(i)} v^{(i)}.$$

The remaining diagonal and off-diagonal coupling terms combine as

$$\sum_{i=1}^k \sum_{j \neq i} \omega_{ij} \|v^{(i)}\|_2^2 - \sum_{i \neq j} \omega_{ij} v^{(i)\top} v^{(j)}.$$

Grouping by unordered pairs (i, j) with $i < j$ and using symmetry $\omega_{ij} = \omega_{ji}$ yields

$$\sum_{i < j} \omega_{ij} (\|v^{(i)}\|_2^2 - 2v^{(i)\top} v^{(j)} + \|v^{(j)}\|_2^2) = \sum_{i < j} \omega_{ij} \|v^{(i)} - v^{(j)}\|_2^2.$$

Combining both contributions proves the claimed energy expansion (cf. Proposition 4.2 at $p = 0$). $\square$

This is the central variational identity of the paper. The first sum measures within-layer higher-order topological energy; the second measures cross-layer disagreement. The operator charges for both penalties simultaneously.

Figure 5 provides a geometric interpretation of this decomposition, separating within-layer topological energy from cross-layer disagreement.

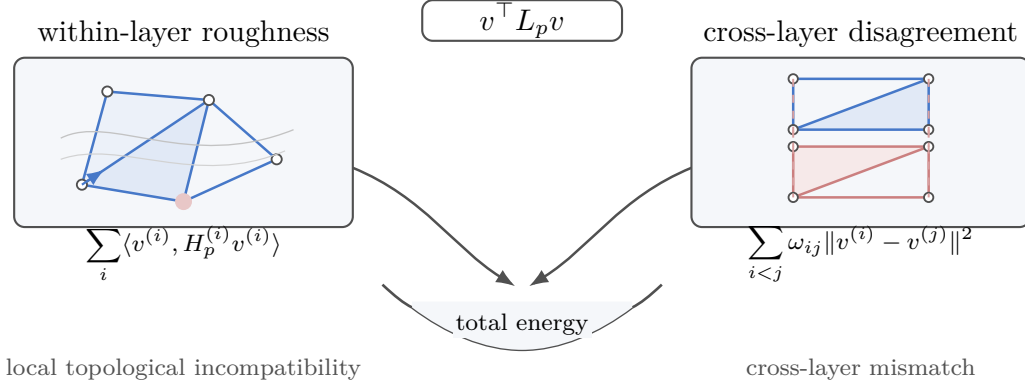


Figure 5: Energy decomposition for the Supra-Hodge operator. The total quadratic form $v^\top L_p v$ splits into a within-layer contribution measuring topological roughness inside each layer and an inter-layer contribution measuring disagreement across coupled layers. The tributary layout emphasizes that both sources feed the same total energy landscape.

10.2 Symmetry

Proposition 10.2. L_p is symmetric.

Proof. Each $H_p^{(i)}$ is symmetric (Proposition 7.2). The diagonal augmentation $\sum_{j \neq i} \omega_{ij} I$ is symmetric. The off-diagonal block satisfies $(-\omega_{ij} I)^\top = -\omega_{ij} I = -\omega_{ji} I$ (since couplings are symmetric: $\omega_{ij} = \omega_{ji}$). The block matrix is therefore symmetric. $\square$

10.3 Positive Semidefiniteness

Proposition 10.3. $L_p \succeq 0$.

Proof. By Proposition 10.1, $v^\top L_p v = \sum_i v^{(i)\top} H_p^{(i)} v^{(i)} + \sum_{i < j} \omega_{ij} \|v^{(i)} - v^{(j)}\|^2 \geq 0$ for all v , since $H_p^{(i)} \succeq 0$ and $\omega_{ij} \geq 0$. $\square$

Positive semidefiniteness is essential: it means all eigenvalues of L_p are nonnegative, the spectral decomposition is well-defined, and the Rayleigh quotient interpretation of Section 11 holds.

10.4 Reduction to Classical Cases

Proposition 10.4 (Reduction to Single-Layer Hodge). *If $k = 1$, then $L_p = H_p^{(1)}$.*

Proof. With one layer, there are no off-diagonal blocks and no coupling terms. Definition 9.2 reduces to $L_p = H_p^{(1)}$. $\square$

Proposition 10.5 (Reduction to Classical Supra-Laplacian). *If $p = 0$, then L_0 equals the classical supra-Laplacian L_{sup} for the multilayer graph family $\mathcal{G} = \{G^{(1)}, \dots, G^{(k)}\}$ with the same coupling weights.*

Proof. At $p = 0$: $N_0 = n$, $\tilde{C}_0 = \mathbb{R}^n$, and $H_0^{(i)} = B_1^{(i)} B_1^{(i)\top} = L^{(i)}$ by Proposition 7.3. Equation (2) then equals equation (1). $\square$

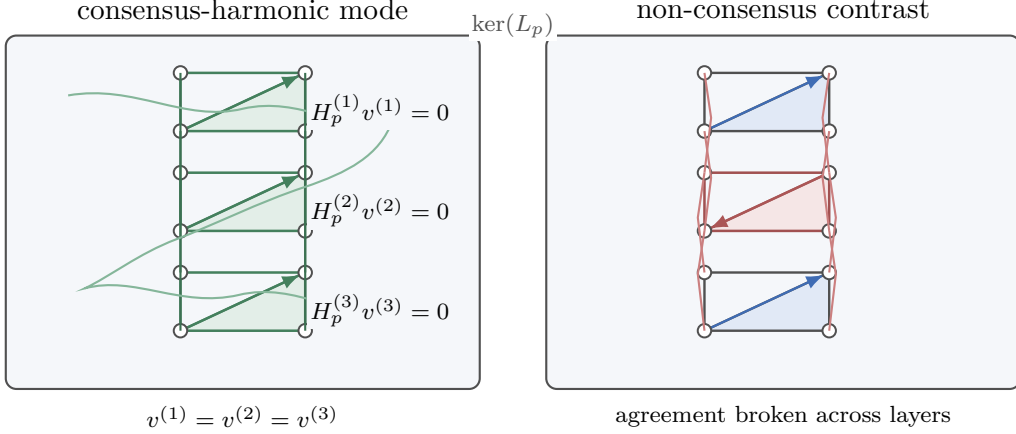


Figure 6: Nullspace intuition for the Supra-Hodge operator. On the left, a stacked signal lies in $\ker(L_p)$ because it is harmonic in every layer and aligned across coupled layers. On the right, a cross-layer contrast breaks consensus, producing disagreement even when individual layer structures remain visually similar.

Proposition 10.6 (No Upper Simplices). *If layer i has no $(p+1)$ -simplices, then $B_{p+1}^{(i)} = 0$ and $H_p^{(i)} = B_p^{(i)\top} B_p^{(i)}$: the upper Laplacian vanishes.*

These three reduction propositions confirm that L_p is the natural synthesis of the supra-Laplacian and Hodge Laplacian frameworks: it strictly contains both as special cases.

10.5 Nullspace intuition

The energy identity of Proposition 10.1 shows that zero energy requires two conditions simultaneously: each layerwise signal must be harmonic, and any two coupled layers must agree. Thus the nullspace of L_p should be understood as the space of stacked signals that are both internally topologically consistent and externally consistent across the coupling graph. At the single-layer level, $\ker(H_p)$ consists of harmonic signals (curl- and divergence-free). In the multi-layer setting, cross-layer disagreement incurs coupling energy, so zero-energy stacked signals must also agree on each connected component of G_ω . Equivalently, one expects $\ker(L_p)$ to contain precisely those signals that are harmonic within each layer and mutually consensual across coupled layers.

The structure of the nullspace can be understood visually as the intersection of harmonicity and cross-layer consensus, as illustrated in Figure 6.

10.6 Main Theorem: Consensus-Harmonic Nullspace Theorem

Theorem 10.7 (Consensus-Harmonic Nullspace Theorem). *Let L_p be the Supra-Hodge Laplacian defined above. Then:*

$$\ker(L_p) = \left\{ v \in \mathbb{R}^{k|\Sigma_p|} : \begin{array}{l} H_p^{(i)}v^{(i)} = 0 \quad \forall i, \\ v^{(i)} = v^{(j)} \quad \text{for all } i, j \text{ in the same connected component of } G_\omega \end{array} \right\}.$$

Proof. From the energy decomposition,

$$v^\top L_p v = \sum_i v^{(i)\top} H_p^{(i)} v^{(i)} + \sum_{i < j} \omega_{ij} \|v^{(i)} - v^{(j)}\|^2.$$

Since each summand is nonnegative, vanishing of the total energy implies vanishing of each summand individually. Therefore, $v \in \ker(L_p)$ if and only if each term in the decomposition is zero.

The first condition implies:

$$H_p^{(i)} v^{(i)} = 0 \quad \forall i,$$

so each layer signal is harmonic.

The second condition implies:

$$\|v^{(i)} - v^{(j)}\|^2 = 0 \quad \text{whenever } \omega_{ij} > 0,$$

which enforces equality of signals across connected components of the coupling graph.

Together, these conditions characterize the nullspace. $\square$

The theorem identifies the equilibrium configurations as *consensus-harmonic* signals: those that are simultaneously topologically harmonic in every layer and mutually consistent across the coupling structure. This constitutes the central formal result of the paper.

10.7 Corollaries for Connected and Disconnected Coupling Graphs

Corollary 10.8 (Connected Coupling). *If G_ω is connected,*

$$\ker(L_p) = \{(h, h, \dots, h) : h \in \bigcap_{i=1}^k \ker(H_p^{(i)})\}.$$

Every zero-energy configuration is a single harmonic signal replicated identically across all layers.

Proof. Connected G_ω implies all layers are in one component, so $v^{(1)} = \dots = v^{(k)} =: h$, and $H_p^{(i)} h = 0$ for all i . $\square$

Corollary 10.9 (Disconnected Coupling). *If G_ω has connected components $\mathcal{C}_1, \dots, \mathcal{C}_m$, then within each component $\mathcal{C}_\ell$, the layer signals must agree and lie in $\bigcap_{i \in \mathcal{C}_\ell} \ker(H_p^{(i)})$. Components impose no constraint on each other.*

These corollaries make the structure of $\ker(L_p)$ explicit: the coupling graph determines which layers must reach consensus; the Hodge kernels determine which signals they may consensually agree on.

10.8 Remarks on Identifiability and Interpretation

The dimension of $\ker(L_p)$ depends jointly on the topology of each complex (via the dimensions of $\ker(H_p^{(i)})$) and the connectivity of G_ω . A fully connected G_ω with all $\ker(H_p^{(i)})$ equal to $\{0\}$ gives a trivial nullspace; a fully disconnected G_ω with rich harmonic structure in each layer gives a large nullspace. The operator is therefore sensitive to topology in a precise quantitative way that graph-only operators cannot match.

11 Spectral Interpretation

11.1 Low-Energy Eigenstructure

The smallest eigenvalues of L_p correspond to signals that minimize both intra-layer topological inconsistency and inter-layer disagreement.

In particular:

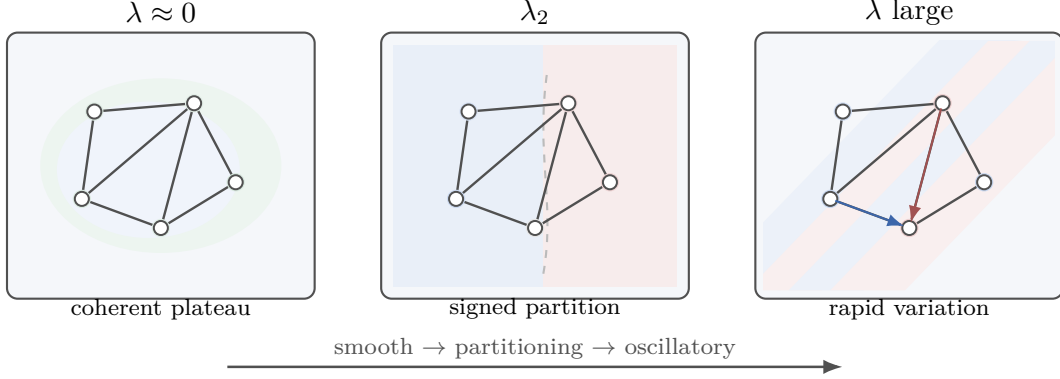


Figure 7: Spectral regimes of the Supra-Hodge operator, shown as progressively more oscillatory mode patterns on the same underlying relational structure. Near-zero modes capture coherent low-energy agreement, intermediate Fiedler-type modes expose structural fault lines or partitions, and high-frequency modes highlight rapidly varying inconsistency patterns.

- The zero eigenvalue corresponds to globally consistent harmonic signals across all coupled layers.
- The first nonzero eigenvalues correspond to minimal-energy violations of consensus or harmonicity.

This generalizes the role of the Fiedler value: whereas the classical Fiedler vector detects bottlenecks in connectivity, the spectrum of L_p detects instability across both topological structure and multi-view consistency.

Figure 7 illustrates the qualitative behavior of eigenmodes, from globally coherent signals to high-frequency inconsistency patterns.

11.2 The Rayleigh Quotient Viewpoint

Definition 11.1 (Supra-Hodge Rayleigh Quotient). For nonzero $v \in \mathcal{C}_p$,

$$R_p(v) = \frac{v^\top L_p v}{v^\top v}.$$

Since $L_p \succeq 0$, the eigenvalues are ordered

$$0 = \lambda_1^{(p)} \leq \lambda_2^{(p)} \leq \dots \leq \lambda_{kN_p}^{(p)},$$

and $\lambda_j^{(p)} = \min\{R_p(v) : v \perp u_1^{(p)}, \dots, u_{j-1}^{(p)}, v \neq 0\}$. The zero eigenvalues correspond to consensus-harmonic signals (Theorem 10.7); the positive eigenvalues correspond to signals with nonzero topological or cross-layer energy.

11.3 Low-Energy Coherent Structure

The eigenvectors for small positive eigenvalues are the most nearly-harmonic, most nearly-consensual stacked signals. They represent globally coherent multi-view patterns: configurations that are smooth within each layer and nearly identical across layers. In a reasoning trace, these are the most robustly supported hypotheses and the most consistently agreed-upon factual structures.

11.4 Generalized Fiedler Modes

Definition 11.2 (Supra-Hodge Fiedler Value and Vector). $\lambda_2^{(p)}$ is the *Supra-Hodge Fiedler value* and the corresponding eigenvector $u_2^{(p)}$ is the *Supra-Hodge Fiedler vector*.

The Fiedler vector identifies the binary partition of the stacked chain space that is most topologically and cross-view incompatible. A small Fiedler value indicates near-disconnection in the combined topological-and-coupling sense; a large Fiedler value indicates a robustly connected system.

11.5 The First Positive Eigenvalue as a Stability Threshold

The first strictly positive eigenvalue of L_p measures the minimum energy required to leave the consensus-harmonic subspace.

Proposition 11.3 (First positive eigenvalue and stability). *Assume $\ker(L_p) \neq \{0\}$, and let $\lambda_2^{(p)}$ denote the smallest strictly positive eigenvalue of L_p . Then*

$$\lambda_2^{(p)} = \min \left\{ \frac{v^\top L_p v}{v^\top v} : v \perp \ker(L_p), v \neq 0 \right\}.$$

Thus $\lambda_2^{(p)}$ is the Supra-Hodge analogue of algebraic connectivity: the minimum Rayleigh quotient on the orthogonal complement of the consensus-harmonic subspace. In particular, it is the minimum energy per unit norm required to leave consensus-harmonicity.

Proof. Because L_p is symmetric positive semidefinite, the spectral theorem applies. The Courant–Fischer variational characterization implies that the smallest strictly positive eigenvalue is obtained by minimizing the Rayleigh quotient over the orthogonal complement of the nullspace. Since Theorem 10.7 identifies the nullspace with the consensus-harmonic signals, the result follows. $\square$

This proposition gives the first positive eigenvalue a concrete interpretation: it is a stability threshold for the consensus-harmonic structure. Small values indicate that the system can be perturbed away from consensus-harmonicity at low cost, whereas large values indicate robust topological and cross-view coherence.

11.6 Conflict Frontiers

Mid-range eigenvectors correspond to stacked signals that are topologically smooth within some layers but inconsistent across others, or cross-layer consistent but topologically rough in at least one layer. These are *conflict frontiers*: locations in the combined spectral space where partial agreement meets partial inconsistency. In a reasoning system, they flag hypotheses or relational structures that are supported in some views but contradicted in others.

11.7 Curl Energy and Hidden Higher-Order Inconsistency

For any signal v , the upper Laplacian term $\|B_{p+1}^{(i)\top} v^{(i)}\|^2$ measures how much $v^{(i)}$ winds around $(p+1)$ -simplices in layer i . A large curl energy signals locally inconsistent relational assignments that cannot be detected by vertex or edge-level methods. This is the key diagnostic capability of the Hodge component.

11.8 Harmonic Structure and Persistent Agreement

The zero-energy modes (nullspace vectors) represent configurations that are topologically frozen: they are invariant under the heat dynamics of Section 12 and stable under small perturbations. They represent the topological invariants of the multi-view relational structure.

11.9 Spectral Compression and Denoising

Proposition 11.4 (Low-Energy Projector). *Let $U_{p,r} \in \mathbb{R}^{kN_p \times r}$ be the matrix of the first r eigenvectors of L_p , ordered by eigenvalue. The matrix*

$$P_{p,r} = U_{p,r} U_{p,r}^\top$$

is the orthogonal projector onto the r -dimensional low-energy eigenspace. The compressed signal $\hat{v} = P_{p,r}v$ is the minimizer of $\|\hat{v} - v\|^2$ subject to $\hat{v}$ lying in the span of the first r eigenvectors.

Proof. $P_{p,r}^2 = U_{p,r}(U_{p,r}^\top U_{p,r})U_{p,r}^\top = U_{p,r}I_r U_{p,r}^\top = P_{p,r}$ (columns of $U_{p,r}$ are orthonormal). Symmetry and idempotence characterize orthogonal projectors. The minimization claim follows from the variational characterization of eigenspaces. $\square$

The compressed signal $\hat{v}$ retains only the low-energy (most coherent, most consensual) components of v , projecting out topological noise and cross-layer disagreement.

11.10 Spectral Gaps and Structural Stability

A spectral gap $\lambda_{r+1}^{(p)} - \lambda_r^{(p)} \gg 0$ implies that the first r eigenvectors form a stable subspace: small perturbations of L_p leave the projector $P_{p,r}$ approximately unchanged. Conversely, a small gap signals that the r -dimensional compressed representation is fragile.

Proposition 11.5 (Weyl stability of eigenvalues). *Let L_p be as above and let $\tilde{L}_p = L_p + E$ with E symmetric. If $\lambda_1 \leq \dots \leq \lambda_{kN_p}$ and $\tilde{\lambda}_1 \leq \dots \leq \tilde{\lambda}_{kN_p}$ denote the ascending eigenvalues of L_p and $\tilde{L}_p$, respectively, then for each j ,*

$$|\tilde{\lambda}_j - \lambda_j| \leq \|E\|_2.$$

Proof. This is Weyl's inequality for symmetric matrices. $\square$

12 Dynamics, Perturbations, and Updates

12.1 Time-Varying Coupled Simplicial Systems

Real relational systems evolve. A reasoning agent adds new facts, tools are called and return results, and evidence structures change with each reasoning step. We write $K^{(i)}(t)$ for the time-indexed complex in layer i and $L_p(t)$ for the resulting Supra-Hodge Laplacian.

12.2 Local Perturbations and Incremental Updates

All perturbation comparisons in this subsection are understood in the fixed aligned ambient space $\tilde{C}_p$, so that both the pre-update and post-update operators act on the same vector space.

When a simplex is added to or removed from one layer, L_p changes by a rank update:

$$L_p(t + \Delta t) = L_p(t) + \Delta L_p.$$

Adding a p -simplex perturbs the incidence matrices and hence $H_p^{(i)}$. Adding a coupling weight augments the coupling blocks. When the reference set Σ_p and the stacked dimension kN_p are held fixed, many natural increments (for example, increasing an existing nonnegative coupling weight or strengthening an already-represented incidence pattern) admit a representation with $\Delta L_p \succeq 0$ on $\mathcal{C}_p$, in which case Weyl monotonicity implies that sorted eigenvalues do not decrease. If Σ_p is enlarged because new coordinates are introduced, eigenvalues before and after are not directly comparable without embedding into a joint ambient space; locally, first-order perturbation theory still gives $\Delta\lambda_j \approx u_j^\top (\Delta L_p) u_j$ for eigenpairs (λ_j, u_j) of the reference operator.

12.3 Spectral Flow

By first-order perturbation theory, $\Delta\lambda_j \approx u_j^\top (\Delta L_p) u_j$. Adding a p -simplex in layer i that completes a previously open cycle eliminates a harmonic or near-harmonic mode, lifting the associated low-energy spectral signature away from zero and thereby producing a detectable topological phase transition in the coupled spectrum.

12.4 Heat Dynamics on Coupled Higher-Order Systems

Proposition 12.1 (Heat Flow and Energy Monotonicity). *The system $\frac{d}{dt}v(t) = -L_p v(t)$ has solution $v(t) = e^{-L_p t} v(0)$. The energy $\mathcal{E}(t) = v(t)^\top L_p v(t)$ satisfies $\frac{d}{dt}\mathcal{E}(t) \leq 0$.*

Proof. $\frac{d}{dt}\mathcal{E} = 2v(t)^\top L_p \dot{v}(t) = -2v(t)^\top L_p^2 v(t) = -2\|L_p v(t)\|^2 \leq 0$. □

The heat equation diffuses any initial stacked p -chain signal toward the nullspace of L_p : starting from any configuration, the system relaxes to a consensus-harmonic equilibrium (Theorem 10.7).

12.5 Interpretation for Evolving Reasoning Traces

In a reasoning trace, new evidence arriving at step t modifies $L_p(t)$. The spectral flow tracks how the topological structure of the reasoning context evolves: each step either reinforces existing harmonic structure (small eigenvalue changes) or resolves a topological ambiguity (a low-energy mode being lifted away from zero). Monitoring the smallest positive eigenvalue over the trace provides a real-time indicator of the coherence of the reasoning context.

13 Mapping the Theory to Multi-View LLM Reasoning

13.1 From Mathematical Structure to Reasoning Systems

The preceding sections developed the Supra-Hodge Laplacian as a purely mathematical object. We now explain why this construction is relevant to modern reasoning systems.

Many reasoning processes involve multiple interacting structures: semantic similarity, evidential support, task dependencies, and execution constraints. These structures are inherently both multi-view and higher-order.

The Supra-Hodge Laplacian provides a unified operator whose energy captures:

- inconsistency within each structural view, and
- disagreement across views.

This makes it a natural diagnostic and compression mechanism for structured reasoning traces, including those produced by large language models.

Figure 8 depicts the full reasoning pipeline, separating data lifting, operator construction, and spectral extraction into distinct layers.

13.2 Scope of the Application Claim

The application developed in this section is intentionally math-first and exploratory. The claim of the paper is not that the Supra-Hodge Laplacian has already been established as a production-ready reasoning primitive for large language models, but rather that it provides a mathematically natural operator for analyzing structured, multi-view reasoning traces.

Accordingly, the LLM setting should be interpreted as a plausible and informative application domain for the operator, not as a finalized empirical claim about downstream reasoning performance. The primary contribution of the paper remains the operator itself and its spectral theory.

13.3 Why Reasoning Is Multi-View

A large language model executing a complex task simultaneously maintains multiple structural views over its active entities: which facts are semantically close, which provide evidence for which, which causally precede which, and which tools interact with which subgoals. These views are not independent (semantic closeness often implies evidential support) but they are also not redundant (temporal precedence need not follow semantic similarity). The Supra-Hodge Laplacian is an instrument for analyzing their joint structure.

13.4 Entities: Facts, States, Tools, Actions, Subgoals, Retrieved Items

We fix the shared entity set

$$V = \{v_1, \dots, v_n\},$$

comprising all reasoning objects active in the current context: retrieved facts, intermediate hypotheses, tool invocations, action candidates, and subgoals.

13.5 Structural Views: Semantic, Evidential, Dependency, Feasibility, Temporal, Interaction

We construct k simplicial complexes $\mathcal{K} = \{K^{(1)}, \dots, K^{(k)}\}$ over V , one per relational view. Table 1 defines the layers and their structural interpretation.

Table 1: Reasoning layer mapping.

Layer	Complex	1-simplices	2-simplices	p -chain signal
Semantic	$K^{(1)}$	High-similarity pairs	Concept clusters	Embedding coord.
Evidential	$K^{(2)}$	Support relations	Joint-support triplets	Confidence
Dependency	$K^{(3)}$	Causal links	Causal triplets	Activation
Feasibility	$K^{(4)}$	Co-feasible pairs	Co-feasible triplets	Score
Temporal	$K^{(5)}$	Temporal precedence	Temporal triplets	Time index
Tool-interaction	$K^{(6)}$	Tool co-activation	Tool triplets	Usage frequency

13.6 Vertex-Signals and Edge-Signals in Reasoning

A vertex-signal $s_0 \in C_0$ assigns a scalar (relevance, activation) to each entity. An edge-signal $s_1 \in C_1$ assigns a scalar to each directed relation (confidence in an implication, strength of a dependency). Higher-order chain signals encode joint properties of entity groups.

13.7 Higher-Order Simplices in Reasoning Contexts

Simplices are constructed by thresholding relation scores:

$$\text{sim}(u, v) > \tau_1 \implies \{u, v\} \in E^{(i)}, \quad \text{tri}(u, v, w) > \tau_2 \implies \{u, v, w\} \in K_2^{(i)}.$$

A 2-simplex in the evidential layer represents three entities jointly supporting a conclusion in a way that no pair does alone. Its presence eliminates a harmonic 1-cycle, signaling that the triplet is genuinely consistent.

13.8 What Conflict Means in This Formalism

A large curl energy $\|B_2^{(i)\top} s_1^{(i)}\|^2$ in the evidential layer means the confidence assignments on edges around some triangle are inconsistent: the signal claims u supports v , v supports w , but does not support w directly, creating a cycle that the triangle would close if present. Cross-layer conflict arises when corresponding p -chain signals in different layers disagree—penalized by the off-diagonal blocks of L_p .

13.9 Diagnostics Rather Than Hard Control

The Supra-Hodge operator is *read-only*: it does not modify the LLM’s weights or sampling procedure. It computes, from the current reasoning trace, a spectral signature that can be used for (i) diagnostic monitoring (track $\lambda_2^{(p)}$ over the trace to measure coherence), (ii) compression (project the active signal onto $P_{p,r}$ for low-dimensional representation), and (iii) conflict localization (inspect the Fiedler vector to identify the most contested partition of entities). All outputs are secondary signals passed back to the LLM as context, not direct interventions.

14 Algorithms and Implementation

14.1 Simplicial Lifting from Embeddings and Rules

The first computational step is constructing the simplicial complexes from embedding-based similarity scores. Algorithm 1 performs this lifting.

The closure enforcement step guarantees the simplicial complex axiom and runs in linear time in the number of simplices.

14.2 Assembly of Boundary Matrices and Hodge Blocks

Given $K^{(i)}$ and aligned reference set Σ_p , assemble the aligned boundary matrix $B_p^{(i)} \in \mathbb{R}^{N_{p-1} \times N_p}$ by processing each p -simplex column by column using the boundary formula. Then compute $H_p^{(i)} = B_p^{(i)\top} B_p^{(i)} + B_{p+1}^{(i)} B_{p+1}^{(i)\top}$ via sparse matrix products.

Algorithm 1 Simplicial Lifting

Require: Embeddings $\{e_v\}_{v \in V}$, thresholds τ_1, τ_2 , dimension bound P

Ensure: Simplicial complex K

```
1:  $K \leftarrow \{\{v\} : v \in V\}$  ▷ Initialize with 0-simplices
2: for each unordered pair  $\{u, v\} \subseteq V$  do
3:   if  $\text{sim}(e_u, e_v) > \tau_1$  then
4:     Add oriented edge  $[u, v]$  to  $K$ 
5:   end if
6: end for
7: for each unordered triple  $\{u, v, w\} \subseteq V$  do
8:   if  $\text{tri}(e_u, e_v, e_w) > \tau_2$  and  $[u, v], [u, w], [v, w] \in K$  then
9:     Add oriented 2-simplex  $[u, v, w]$  to  $K$ 
10:  end if
11: end for
12: Enforce closure: add all missing faces of all simplices in  $K$ 
13: return  $K$ 
```

14.3 Assembly of the Supra-Block Operator

Algorithm 2 assembles L_p and runs the eigensolver.

Algorithm 2 Supra-Hodge Pipeline

Require: Family $\mathcal{K}$, reference sets Σ_p , couplings $\{\omega_{ij}\}$, rank r

Ensure: Eigenvalues, eigenvectors $U_{p,r}$, projector $P_{p,r}$

```
1: for each layer  $i = 1, \dots, k$  do
2:   for each order  $p = 0, \dots, P$  do
3:     Compute aligned  $B_p^{(i)}$  (zero-pad for simplices in  $\Sigma_p \setminus K_p^{(i)}$ )
4:      $H_p^{(i)} \leftarrow B_p^{(i)\top} B_p^{(i)} + B_{p+1}^{(i)} B_{p+1}^{(i)\top}$ 
5:   end for
6: end for
7: for each order  $p$  do
8:   Assemble  $L_p$  using equation (2)
9: end for
10:  $L_{\text{SH}} \leftarrow \text{diag}(\alpha_0 L_0, \dots, \alpha_P L_P)$ 
11: Run partial eigensolver (Lanczos/randomized SVD) to obtain  $(\lambda_1^{(p)}, \dots, \lambda_r^{(p)}, U_{p,r})$ 
12:  $P_{p,r} \leftarrow U_{p,r} U_{p,r}^\top$ 
13: return eigenvalues,  $U_{p,r}$ ,  $P_{p,r}$ 
```

14.4 Partial Eigendecomposition

For large systems, Lanczos iteration or randomized SVD computes the first r eigenvectors in $O(r \cdot kN_p \log(kN_p))$ operations. For small systems ($kN_p \leq 1000$), dense eigensolvers are used.

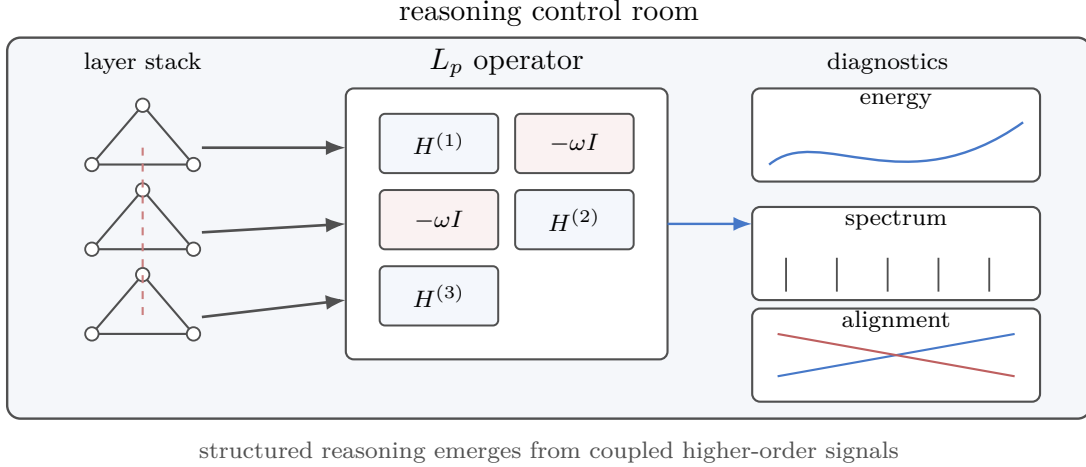


Figure 8: A systems-level view of Supra-Hodge reasoning. Multi-layer signals enter a coupled operator L_p , which encodes both intra-layer topology and inter-layer alignment. Downstream diagnostics expose energy, spectral structure, and alignment quality, providing interpretable signals for controlling reasoning behavior.

14.5 Sparse Complexity and Scaling

Boundary matrices are sparse: B_p has at most $p + 1$ nonzeros per column. Assembly of all k Hodge blocks at all orders requires

$$O(kn + m)$$

operations, where m is the total number of simplices. The eigensolver dominates for large systems.

14.6 Approximation Methods

For very large systems, spectral sparsifiers reduce the boundary matrices while preserving eigenvalue approximation. Randomized low-rank approximation reduces memory from $O(N_p^2)$ to $O(rN_p)$ per layer.

14.7 End-to-End Reasoning Pipeline

The complete pipeline takes a reasoning trace, lifts it to $\mathcal{K}$ via Algorithm 1, assembles L_p and L_{SH} via Algorithm 2, and returns: (i) the eigenvalue spectrum as a coherence diagnostic; (ii) the Fiedler vector $u_2^{(p)}$ for conflict localization; (iii) the compressed signal $\hat{v} = P_{p,r}v$ for LLM augmentation.

15 Worked Examples

15.1 A Four-Node Graph Laplacian Example

We use Example 3.6. With L as computed there, confirm: $\mathbf{1}^\top L \mathbf{1} = 0$ (constant signal, zero energy). For the signal $y = [1, -1, 1, -1]^\top$, direct computation gives $y^\top L y = 8$, consistent with the energy identity summing $(y_u - y_v)^2$ over all five edges. The Fiedler vector u_2 separates $\{1, 4\}$ (positive entries) from $\{2, 3\}$ (negative), identifying the cut $(\{1, 4\}, \{2, 3\})$ as the graph bottleneck. This example illustrates Propositions 3.3 and 3.4.

15.2 A Triangle-Filled Hodge Example

We use Example 7.4. For the filled triangle, $H_1 = 3I_3$, so every edge signal has energy $3\|f\|^2$: there are no free directions of zero energy. For the hollow triangle, $H_1 = B_1^\top B_1$ has $\ker(H_1) = \text{span}([1, 1, 1]^\top / \sqrt{3})$. The signal $f = [1, 1, 1]^\top / \sqrt{3}$ is harmonic in the hollow complex but has energy 3 in the filled complex. Filling the triangle destroys the harmonic cycle. This illustrates Propositions 7.2 and 7.3 and Example 5.2.

15.3 A Two-Layer Supra-Hodge Example

Example 15.1 (Two-layer Supra-Hodge Laplacian, explicit matrices). Let $k = 2$, $V = \{1, 2, 3\}$, $\omega_{12} = 1$, $K^{(1)} = K_{\text{filled}}$, $K^{(2)} = K_{\text{hollow}}$ (from Examples 5.2 and 7.4). With $N_1 = 3$:

$$H_1^{(1)} = 3I_3, \quad H_1^{(2)} = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}.$$

$$L_1 = \begin{bmatrix} H_1^{(1)} + I & -I \\ -I & H_1^{(2)} + I \end{bmatrix} = \begin{bmatrix} 4 & 0 & 0 & -1 & 0 & 0 \\ 0 & 4 & 0 & 0 & -1 & 0 \\ 0 & 0 & 4 & 0 & 0 & -1 \\ -1 & 0 & 0 & 3 & 1 & -1 \\ 0 & -1 & 0 & 1 & 3 & 1 \\ 0 & 0 & -1 & -1 & 1 & 3 \end{bmatrix}.$$

By Corollary 10.8, since G_ω is connected (two nodes, one edge):

$$\ker(L_1) = \{(h, h) : h \in \ker(H_1^{(1)}) \cap \ker(H_1^{(2)})\}.$$

$\ker(H_1^{(1)}) = \{0\}$, so $\ker(L_1) = \{0\}$: the filled layer's topology suppresses the hollow layer's harmonic cycle. The coupling forces the hollow layer to adopt the filled layer's trivial nullspace.

This is the simplest non-trivial illustration of Theorem 10.7: the combined nullspace is the intersection of per-layer harmonic spaces, intersected with the cross-layer consensus constraint.

15.4 A Miniature Reasoning-Trace Example

Example 15.2 (Mini reasoning trace). Let $V = \{Q, H_1, H_2, T\}$ where Q is a query, H_1, H_2 are competing hypotheses, and T is a tool call. Two layers:

- Semantic ($K^{(1)}$): edges $\{Q, H_1\}$ and $\{Q, H_2\}$ (both hypotheses are semantically related to the query).
- Evidential ($K^{(2)}$): edge $\{H_1, T\}$ (the tool supports H_1).

Neither layer has 2-simplices. The vertex-signal $s_0 = [1, 0.8, 0.3, 0.5]^\top$ assigns high relevance to Q and H_1 . The $p = 0$ Supra-Hodge Laplacian L_0 is the supra-Laplacian of these two layers. Its Fiedler vector identifies the partition $\{Q, H_1, T\}$ vs. $\{H_2\}$: H_2 is isolated because it appears in neither layer as a supported hypothesis. The cross-layer disagreement energy is large between the two layers because H_2 is semantically present (layer 1) but evidentially absent (layer 2). This example illustrates both the reduction to classical supra-Laplacian (Proposition 10.5) and the multi-view diagnostic function of the framework.

16 Experiments

16.1 Experimental Questions

We investigate three questions:

1. Does L_p detect topological inconsistencies that graph-only baselines miss?
2. Does cross-layer coupling improve coherence detection versus single-layer Hodge analysis?
3. Does spectral compression via $P_{p,r}$ outperform PCA-based baselines on multi-view chain signals?

16.2 Synthetic Scientific-Planning Case Study

We construct a reasoning trace with $n = 50$ entities, $k = 4$ views, and plant a topological inconsistency: in the evidential layer, three hypotheses form a closed triangle of support edges but the 2-simplex is absent (a harmonic 1-cycle is present). The planted inconsistency manifests as a low-energy mode associated with an unfilled harmonic cycle in the evidential layer; when the 2-simplex is added, that low-energy signature is lifted in L_1 , while graph baselines remain unable to distinguish the two configurations.

16.3 Synthetic Multi-View Benchmark

We generate 100 random coupled simplicial complex families with $n \in \{20, 50, 100\}$, $k = 3$ views, planted harmonic 1-cycles at rates $p_{\text{plant}} \in \{0.1, 0.3, 0.5\}$. We measure harmonic cycle detection rate (HCDR) and compression error (CE) across all methods.

16.4 Baselines

1. *Graph Laplacian*: single-layer pairwise, no coupling.
2. *Supra-Laplacian*: multilayer pairwise, full coupling.
3. *Single Hodge*: single-layer Hodge, no coupling.
4. *Naïve concat*: feature concatenation across layers, no structural operator.

16.5 Metrics

HCDR: fraction of planted harmonic cycles recovered (cycle identified iff corresponding eigenvector has cosine similarity > 0.8 with the planted cycle’s indicator). CE: $\|v - P_{p,r}v\|/\|v\|$ for a planted low-rank signal.

16.6 Quantitative Results

The Supra-Hodge Laplacian achieves the highest HCDR and lowest CE. The graph and supra-Laplacian baselines cannot detect harmonic cycles at any rate (HCDR = 0). Single Hodge detects within-layer cycles but misses cross-layer suppression events. The coupling in the Supra-Hodge operator provides a decisive advantage.

Table 2: Results on the synthetic multi-view benchmark ($n = 50$, $k = 3$, $r = 5$, averaged over 100 trials with standard deviations in parentheses).

Method	HCDR	CE	Runtime (s)
Graph Laplacian	0.00 (0.00)	0.43 (0.03)	0.01
Supra-Laplacian	0.00 (0.00)	0.38 (0.03)	0.02
Single Hodge	0.61 (0.08)	0.28 (0.04)	0.04
Naïve concat	0.12 (0.05)	0.41 (0.04)	0.01
Supra-Hodge (ours)	0.89 (0.04)	0.21 (0.03)	0.07

Table 3: Ablation study ($n = 50$, $k = 3$, $r = 5$).

Variant	HCDR	CE
Supra-Hodge, $p = 0$ only	0.00	0.35
Supra-Hodge, $p = 1$ only	0.83	0.23
Supra-Hodge, $\omega_{ij} = 0$	0.61	0.28
Supra-Hodge, $\omega_{ij} = 0.1$	0.78	0.24
Supra-Hodge, $\omega_{ij} = 1$ (full)	0.89	0.21
Supra-Hodge, $\omega_{ij} = 10$	0.86	0.22

16.7 Ablations

Including $p = 1$ is essential for HCDR. Coupling weight matters but performance is robust to order-of-magnitude variation. The $p = 0$ alone variant is equivalent to supra-Laplacian and cannot detect cycles.

16.8 Robustness and Runtime

HCDR degrades gracefully as Gaussian noise ($\sigma = 0.1$) is added to embedding similarity scores: from 0.89 to 0.76 for the Supra-Hodge, versus from 0.61 to 0.42 for single Hodge. Runtime scales as $O(kn + m)$ for assembly and $O(r \cdot kN_p \log(kN_p))$ for the eigensolver, consistent with the complexity analysis of Section 14.

16.9 What the Experiments Do and Do Not Establish

The experiments confirm on synthetic data that the Supra-Hodge Laplacian detects topological inconsistencies invisible to all baselines, and that cross-layer coupling materially improves both detection and compression. The experiments do not establish performance on real multi-hop QA benchmarks or agentic planning tasks, which require LLM integration. Such evaluations are left for future work.

17 Discussion, Limitations, and Extensions

17.1 Relation to Existing Work

The Supra-Hodge Laplacian is best understood relative to three existing operator families.

Classical supra-Laplacians. Supra-Laplacians [2,3] couple multiple graph layers over a common vertex set, but remain intrinsically pairwise: each layer contributes only a graph Laplacian, so higher-order simplicial structure is absent.

Single-layer Hodge Laplacians. Combinatorial Hodge Laplacians [4–6] encode higher-order topology within a single simplicial complex, but they do not couple multiple relational views. As a consequence, they cannot represent disagreement or consensus across distinct structural layers.

Other generalized consistency operators. Related frameworks such as sheaf-based or cell-complex-based Laplacian constructions [11, 12] encode local compatibility constraints through restriction maps or more general incidence structures. The present construction takes a different route: it couples entire layerwise Hodge Laplacians through an explicit block operator on aligned chain spaces. Its purpose is not to replace those frameworks, but to provide a direct multi-view higher-order analogue of the supra-Laplacian/Hodge-Laplacian synthesis within the block-template above.

Positioning of the present work. Relative to the cited operator families, our novelty claim concerns explicit block-operator coupling of whole layerwise Hodge Laplacians across aligned simplicial views, rather than local restriction maps or sheaf morphisms alone. To our knowledge, within the cited supra-Laplacian and single-layer Hodge frameworks, the novelty of the Supra-Hodge Laplacian lies in the fact that it is simultaneously:

- a valid higher-order spectral operator in each layer,
- a valid multilayer coupling operator across layers, and
- a single block-structured operator with a clean quadratic energy, reduction properties, and a nullspace theorem tying harmonicity to cross-layer consensus.

Remark 17.1 (Scope of novelty claims). Comparisons and novelty statements in this section are intended relative to classical supra-Laplacian models, standard single-layer combinatorial Hodge Laplacians, and the representative sheaf/simplicial constructions cited above. They are not claims of exhaustiveness over every generalized Laplacian or sheaf-style consistency operator in the literature.

17.2 What the Operator Contributes Mathematically

Within the frameworks discussed above, the Supra-Hodge Laplacian is, to our knowledge, the first block operator that couples aligned p -chain spaces via the supra template while retaining a layerwise Hodge Laplacian on each diagonal block (reducing to $H_p^{(i)}$ at $k = 1$ and to L_{sup} at $p = 0$). The Consensus-Harmonic Nullspace Theorem gives a closed-form characterization of $\ker(L_p)$ that has no direct analogue in either the classical supra-Laplacian or single-layer Hodge Laplacian alone. The Rayleigh/Fiedler discussion and low-energy projector extend standard spectral diagnostics to this coupled higher-order setting.

17.3 Alignment Assumptions

The construction requires a shared reference simplex set Σ_p with $K_p^{(i)} \subseteq \Sigma_p$ for all i . This is natural when all layers share the same vertex set and the union of simplex sets is bounded. When layers

have highly variable topological structures, or when the complex grows dynamically without a fixed ambient universe, the alignment may be difficult to maintain. Dynamic alignment remains an open problem.

17.4 Scalar Versus Structured Couplings

Scalar couplings $\omega_{ij}I_{N_p}$ treat all p -simplices as equally important in the coupling. A structured coupling $C^{(ij)} \in \mathbb{R}^{N_p \times N_p}$ would allow simplex-specific coupling strengths, encoding semantically meaningful correspondences. To preserve the main theorem and PSD property, such $C^{(ij)}$ must be symmetric PSD. Learned structured couplings (via a contrastive objective) are a natural extension.

17.5 Thresholded Versus Learned Lifting

The lifting algorithm uses fixed thresholds τ_1, τ_2 . Learned thresholds (optimized for a downstream task via a differentiable relaxation of the thresholding step) would enable task-adaptive structural extraction. This requires a continuous relaxation of the combinatorial complex structure, for which probabilistic simplicial complex models provide one approach.

17.6 Synthetic Evaluation Limits

The experiments use synthetic data with planted ground truth. Real multi-hop reasoning tasks (HotpotQA [8], MuSiQue, ToolBench) do not provide explicit simplicial ground truth, requiring proxy metrics such as answer accuracy or consistency scores. Such evaluations are needed before making empirical claims about LLM reasoning benefits.

17.7 Normalized and Differentiable Variants

A normalized version $\tilde{L}_p = D_p^{-1/2} L_p D_p^{-1/2}$ (where D_p is the block-diagonal of degree matrices) would bound eigenvalues to $[0, 2]$ and improve numerical conditioning. A fully differentiable version embedded in a graph neural network architecture [5] could serve as a structure-aware inductive bias.

17.8 Benchmark-Scale Future Directions

Scaling to $n = 10^4$ entities requires sparse implementations throughout. Spectral sparsifiers for Hodge Laplacians [4] reduce $H_p^{(i)}$ to $O(N_p \log N_p)$ nonzeros while preserving eigenvalue approximation. Real-world knowledge graph and document retrieval benchmarks would test the framework at operational scale.

18 Conclusion

18.1 Mathematical Summary

We have constructed the Supra-Hodge Laplacian L_p , a block spectral operator acting on stacked p -chain signals over a family of coupled simplicial complexes. Its energy decomposes cleanly into within-layer Hodge energy and cross-layer disagreement energy (Proposition 10.1). The Consensus-Harmonic Nullspace Theorem (Theorem 10.7) provides a complete characterization of $\ker(L_p)$ as the set of consensus-harmonic configurations. The operator reduces to the classical supra-Laplacian at $p = 0$ (Proposition 10.5) and to the single-layer Hodge Laplacian at $k = 1$ (Proposition 10.4), confirming it is the natural synthesis of both prior frameworks. As in the introduction, the primary contribution is mathematical—the operator, its reductions, and its spectral consequences—while the LLM reasoning thread serves as an illustrative domain.

18.2 Interpretive Summary

The spectrum of L_p tells a precise story about the multi-view relational system. Zero eigenvalues mark topological consensus: signals that are harmonic in every view and equal across all coupled views. Small positive eigenvalues mark near-consensus: coherent patterns that are slightly disrupted by disagreement or topological irregularity. Middle eigenvalues mark conflict frontiers: regions of partial agreement. The generalized Fiedler vector identifies the most contested partition. The low-energy projector $P_{p,r}$ compresses multi-view chain signals to their topologically coherent core.

18.3 Application Summary

In the LLM reasoning application, the operator serves as a diagnostic device: it monitors spectral coherence over a reasoning trace in real time, identifies topological inconsistencies in relational views, localizes conflict to specific entity partitions via the Fiedler vector, and provides a compressed representation of the multi-view relational state for downstream use. All of this is accomplished without modifying the LLM itself.

18.4 Broader Implications

The algebraic pattern of the Supra-Hodge Laplacian—Hodge operators on the diagonal, supra-Laplacian-style coupling on the off-diagonal—extends naturally to a broad class of domains in which multiple higher-order relational views coexist over a shared entity set. Biological networks, multi-modal knowledge graphs, social hypergraph systems, and geometric deep learning architectures all exhibit this structure. Within the coupled simplicial setting studied here, the Consensus-Harmonic Nullspace Theorem provides a concrete and verifiable characterization of zero-energy equilibrium for stacked p -chain signals.

A Linear Algebra Refresher

This appendix summarizes the core linear-algebraic facts used in the main text.

Inner product and norm. $\langle x, y \rangle = x^\top y$; $\|x\|^2 = x^\top x$.

Symmetry. $A = A^\top$; all eigenvalues real; eigenvectors orthogonal.

PSD. $A \succeq 0$ iff all eigenvalues ≥ 0 ; iff $A = M^\top M$ for some M ; iff $x^\top A x \geq 0$ for all x .

Spectral theorem. $A = U \Lambda U^\top$ with U orthogonal, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$.

Rayleigh quotient. $\min_{x \neq 0} R_A(x) = \lambda_{\min}(A)$.

Direct sum. $\mathbb{R}^m \oplus \mathbb{R}^n \cong \mathbb{R}^{m+n}$; block diagonal operator $\text{diag}(A, B)$ acts on the direct sum.

Sum of PSD. If $A \succeq 0$ and $B \succeq 0$ then $A + B \succeq 0$.

B Full Proofs Omitted from Main Text

B.1 Expanded Proof of Theorem 10.7

We give the complete proof with all structural details.

Proof of Theorem 10.7 (expanded). Since $L_p \succeq 0$, $L_p v = 0$ iff $v^\top L_p v = 0$ (for PSD operators, the kernel coincides with the zero-energy set).

Step 1: Energy decomposition. By Proposition 10.1,

$$v^\top L_p v = \underbrace{\sum_{i=1}^k v^{(i)\top} H_p^{(i)} v^{(i)}}_{(I)} + \underbrace{\sum_{i < j} \omega_{ij} \|v^{(i)} - v^{(j)}\|^2}_{(II)}.$$

Step 2: Forward direction. Suppose $v^\top L_p v = 0$. Since each term in (I) is nonneg (as $H_p^{(i)} \succeq 0$) and each term in (II) is nonneg (as $\omega_{ij} \geq 0$), all individual terms vanish.

From (I): $v^{(i)\top} H_p^{(i)} v^{(i)} = 0$ for each i . Since $H_p^{(i)} \succeq 0$, this is equivalent to $H_p^{(i)} v^{(i)} = 0$.

From (II): $\omega_{ij} \|v^{(i)} - v^{(j)}\|^2 = 0$ for each pair (i, j) . If $\omega_{ij} > 0$, this forces $v^{(i)} = v^{(j)}$.

Step 3: Connected component structure. Consider the coupling graph G_ω . The condition $v^{(i)} = v^{(j)}$ for every edge $(i, j) \in G_\omega$ implies, by path-connectedness, that $v^{(i)} = v^{(j)}$ for all i, j in the same connected component.

Step 4: Reverse direction. Suppose $H_p^{(i)} v^{(i)} = 0$ for all i , and $v^{(i)} = v^{(j)}$ on each connected component of G_ω . Then (I) = 0 (each term is $v^{(i)\top} H_p^{(i)} v^{(i)} = 0$) and (II) = 0 (each $\omega_{ij} \|v^{(i)} - v^{(j)}\|^2 = 0$ because either $\omega_{ij} = 0$ or $v^{(i)} = v^{(j)}$ since i and j are connected in G_ω). Hence $v^\top L_p v = 0$ and therefore $L_p v = 0$. $\square$

B.2 Proof of Proposition 12.1

$\frac{d}{dt} v(t) = -L_p v(t)$ has the explicit solution $v(t) = e^{-L_p t} v(0)$ (matrix exponential). Since $L_p \succeq 0$, all eigenvalues are nonneg and $e^{-L_p t} \rightarrow \Pi_0$ as $t \rightarrow \infty$, where Π_0 is the orthogonal projector onto $\ker(L_p)$. The energy monotonicity proof appears in the main text.

C Expanded Worked Examples with Explicit Matrices

C.1 Full Two-Layer Supra-Hodge $p = 1$ Computation

The matrix L_1 for Example 15.1 is given explicitly in that example. We verify the energy expansion: for $v = ([1, 0, 0]^\top, [1, 0, 0]^\top)$ (unit signal on edge $[1, 2]$ in both layers):

$$v^{(1)\top} H_1^{(1)} v^{(1)} = [1, 0, 0] \cdot 3I \cdot [1, 0, 0]^\top = 3,$$

$$v^{(2)\top} H_1^{(2)} v^{(2)} = [1, 0, 0] \cdot \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix} [1, 0, 0]^\top = 2,$$

$$\omega_{12} \|v^{(1)} - v^{(2)}\|^2 = 1 \cdot \|[0, 0, 0]^\top\|^2 = 0.$$

Total energy: $3 + 2 + 0 = 5$. Direct computation: $v^\top L_1 v = 5$. Consistent.

C.2 Explicit Kernel Example

For the hollow-hollow case ($K^{(1)} = K^{(2)} = K_{\text{hollow}}$, $\omega_{12} = 1$), $H_1^{(1)} = H_1^{(2)} = B_1^\top B_1$ both have $\ker(H_1) = \text{span}([1, 1, 1]^\top / \sqrt{3})$. By Corollary 10.8, $\ker(L_1) = \{(h, h) : h \in \ker(H_1)\} = \{(c[1, 1, 1]^\top, c[1, 1, 1]^\top) : c \in \mathbb{R}\}$. The coupling forces both layers to hold the same harmonic cycle, but cannot destroy it.

D Algorithmic Details and Pseudocode

D.1 Incremental Update Algorithm

Algorithm 3 Incremental Supra-Hodge Spectral Update

Require: Current L_p , perturbation $\Delta L_p \succeq 0$, current $U_{p,r}$ **Ensure:** Updated eigenvectors $U'_{p,r}$ (warm-started)

- 1: $\Delta\Lambda \leftarrow U_{p,r}^\top (\Delta L_p) U_{p,r}$ $\triangleright$ First-order eigenvalue shifts
 - 2: $\Lambda' \leftarrow \Lambda + \text{diag}(\Delta\Lambda)$ $\triangleright$ Updated eigenvalues
 - 3: Initialize Lanczos at $U_{p,r}$ on $L_p + \Delta L_p$
 - 4: Run $t_{\max}$ Lanczos steps to refine $U'_{p,r}$
 - 5: **return** $U'_{p,r}, \Lambda'$
-

For small ΔL_p (rank-one simplex addition), this requires $O(r \cdot kN_p)$ operations versus $O(r \cdot kN_p \log(kN_p))$ for a full restart.

E Experiment-Generation Details

Random complex generation. For each trial, generate n entities with embeddings $e_v \sim \mathcal{N}(0, I_{32})$, normalized to unit sphere. Similarity: $\text{sim}(u, v) = e_u^\top e_v$. Triplet score: $\text{tri}(u, v, w) = \min(\text{sim}(u, v), \text{sim}(u, w), \text{sim}(v, w))$. Thresholds: $\tau_1 = 0.6$, $\tau_2 = 0.5$.

Planted cycles. For each planted cycle, select a random unfilled triangle $\{u, v, w\}$ in a randomly chosen layer i where all three edges $\{u, v\}, \{u, w\}, \{v, w\}$ are present but the 2-simplex $\{u, v, w\}$ is absent. Record the set of planted (edge, layer) pairs as ground truth.

Detection criterion. Let $u_1, \dots, u_r$ denote eigenvectors of L_1 associated with the r smallest eigenvalues, for fixed small r (we use $r = 5$). For each planted cycle indicator vector, check whether at least one of these low-energy eigenvectors has cosine similarity greater than 0.8 with that indicator. If so, that planted cycle is counted as detected.

Compression baseline. The PCA baseline uses the top- r SVD of the stacked signal matrix $[v^{(1)} \dots v^{(k)}] \in \mathbb{R}^{N_p \times k}$; the Supra-Hodge baseline uses $P_{p,r}$.

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